HiCAP: Hierarchical Contrastive Action Priors on a Frozen Vision–Language Backbone for Elastic, Imagination-Aware Driving

TanitAD programme

Design and pre-registration paper
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Abstract

Autonomous-driving policies built on vision–language models (VLMs) inherit world knowledge but decode actions one token at a time, which is bound by memory bandwidth on an in-vehicle computer. Contrastive selectors (CLM, CoVer-VLA) replace generation by a single prefill pass and a similarity search over cached action embeddings, but they treat the action set as flat and small, inherit the marginal-frequency bias of their loss, and neither predict outcomes nor budget their sensors. We describe *HiCAP*, a design that keeps what made CLM work—a frozen backbone, two small trainable heads, cached action embeddings, a calibrated cosine score, a staged curriculum—and adapts it to driving. (i) The action side is a *tree* built from the programme's frozen tactical and strategic token vocabulary, refined by a factored product of geometric paths and speed profiles, so that vocabularies of 10^5–10^6 entries (growth gated on a test) are scored coarse-then-compose (a cost proposition, a beam-miss bound and a measured pruning study are given). (ii) A short chain of prior-encoded rules—kinematic feasibility, state-conditioned admissibility, hysteresis, and conformal keep-sets with a stated recall bound—reduces the candidate set before any expensive computation, and the survivors are fed back to the backbone for candidate-conditioned re-scoring. (iii) We show that the unmasked InfoNCE optimum is a pointwise mutual information whose argmax is biased under a cruise-dominated prior, that CLM's same-action mask changes that optimum, and correct with a fitted log-prior or a soft-target loss. (iv) Imagination is confined to what data can identify and is built in three tiers: outcome-aware soft targets at training time, an analytically integrated forecast of a few non-reactive exogenous descriptors, and a gated contrastive critic whose head disagreement is a veto; a controlled toy attributes most of the apparent gain to the target shape. (v) An elastic-sensing controller chooses cameras and visual-token budgets per tick from vision-only cues (ego cues as an ablation), and an optional audio branch enters the backbone's embedding space as a zero-initialised gated residual whose null is bit-exact, with explicit controls. (vi) The backbone is a VLM chosen from the NVIDIA Cosmos family and its Qwen3-VL relatives, and its size is set by a *latency* criterion, not a parameter count: the command path is a small fast network, the VLM is refreshed asynchronously at 2 Hz off the critical path, and the price of a larger backbone is the age of its cached tokens, not command latency. Arithmetic anchored on NVIDIA-published Jetson Thor rows admits an 8 B-class backbone at three cameras under a stated duty-cycle rule; larger models are brackets outside the measured range and are gated on an on-device measurement. A distilled student remains available with a decision-preservation bound but is optional. We report only what exists: closed-form propositions with proofs, a real-data check that a physical-limit mask removes 78% of a speed-agnostic trajectory set (15% of a speed-normalised one) while retaining all 881/881 logged futures, toy measurements of the prior bias, and a pre-registered protocol with both outcomes committed for every hypothesis. No end-to-end HiCAP model has been trained.

1 Introduction

A driving policy has to do three things that pull in different directions: know a great deal about the world (what a school bus implies, what a siren means), reason at several time scales (which road to take, which manoeuvre to start, how to move the wheel now), and fit into a fixed power and latency envelope. Large vision–language models supply the first; the field's dominant way of using them, autoregressive generation of a plan or trajectory [14, 28, 33], works against the third, because decoding one token requires streaming the model's weights once and is bound by memory bandwidth rather than arithmetic on an edge SoC.

Contrastive Language Models (CLM) showed a different way to use a large frozen model [18]: read the model once (prefill), map the last hidden state and each candidate action through two small MLP heads into one shared 512-dimensional space, and choose the action with the highest cosine score. Action embeddings are computed once and cached, the backbone is never updated, and a three-stage curriculum trains the heads

on cached features. CoVer-VLA transferred the recipe to robot manipulation with a frozen image encoder and numeric action chunks [19]. Both treat the action set as a flat list.

Driving has structure that a flat list wastes. Actions are *hierarchical*: a route-level intent, a manoeuvre (lateral and longitudinal, chosen separately), and a continuous trajectory. Actions are *highly prior-structured*: physics, the current speed, the last plan and the navigation command exclude most of the space before any learning. The candidate set can be very large if it can be searched cheaply. And the input side is not fixed: not every camera or token is worth its cost at every instant, and some decisive cues (a siren) are not visual at all.

Programme context. HiCAP is the vision–language arm of TanitAD, a programme building hierarchical latent world models for driving whose other arms are sub-300 M. For this arm the PI has lifted the size cap and replaced it by a latency criterion (§8.3): the parameter count may be large if the whole system's inference time stays low. The programme's newest reference planner line (anchored diffusion, REF-C) exposes the problem this paper addresses. On the parity lineage, the best candidate in a 256-anchor fan has an ADE@2 s of 0.1640 m while the candidate the planner selects has 0.4714 m[I] (REF-C-XL, 40 validation episodes; registry §4). On the newest lineage, trained on the v7 corpus, the hold-the-current-action control scores 0.2996 m and its extension 0.2874 m, against 0.2975 m and 0.3079 m for the two best trained arms[I] (the PI's Drive copy of the registry, §4.6–4.8, which is not in this repository; 4,823 windows, 141 episodes; neither arm is separated from the controls). In the older arm whose gap to the hold control was largest, 92.2% of that gap was along-track[I], and the newest arm still trails the controls on speed error (0.2919 against 0.2540 m/s) and along-track error (0.2655 against 0.2348 m) while beating them on heading, yaw rate and curvature[I]. Strategic (route-level) quality is UNAVAILABLE ($n = 0$) for the older arms and 0.7708 route accuracy against a 0.6742 majority rate for the newest[I], from a head of 771 parameters whose attribution is itself uncertain. In short, the programme's bottleneck is neither the candidate set nor the lateral path; it is the *decision*, especially the longitudinal one, and it is under-supervised at the tactical and strategic levels. HiCAP is our attempt to build the decision maker properly on a backbone that already knows the world.

Contributions. We (1) define a hierarchical action tree from the programme's frozen tactical/strategic token vocabulary with a factored path-by-profile trajectory vocabulary beneath it, with a retrieval-cost proposition, a beam-miss bound and a real-data pruning study (§4.2, §7, §9); (2) show that a CLM-style contrastive selector is prior-biased and give a corrected, per-level objective (§4.3, Corollary 2); (3) specify prior-encoded pruning with a conformal recall bound and a candidate-conditioned re-scoring stage that returns survivors to the backbone (§4.4); (4) restrict imagination to what data can identify, in three tiers (soft outcome-aware targets, an analytic forecast of exogenous descriptors, a gated critic with a disagreement veto) and show in a toy that most apparent gain is target shape (§5, Proposition 9); (5) add elastic sensing (cameras, tokens, embedding width) and an optional audio branch tokenised into the backbone's space (§6); (6) replace the parameter cap by a latency criterion with two clocks, command latency and token age, give the duty-cycle arithmetic that says which backbone sizes it admits on the target SoC, and keep a distilled student, with a decision-preservation guarantee, as an option that is used only if a measured latency tier fails (§8, Proposition 11); and (7) commit a pre-registered protocol, including negative controls and both outcomes for each hypothesis (§10).

Scope and evidence convention. This is a design and pre-registration paper. **No HiCAP model has been trained.** Everything quantitative is one of: a closed-form statement with a proof; a measurement on committed data using only inference-time quantities; a toy simulation; arithmetic; or a number inherited from a programme document. Each number carries an evidence tag: [M] measured by us with an artifact in the repository, [P] published and cited, [I] inherited from a programme document and not re-verified here, [E] closed-form estimate (no timing), [H] hypothesis. Where two programme documents disagree, the model registry is the tie-breaker and the conflict is reported (§12).

2 State of the art

Planning heads: from regression to scored candidate sets. End-to-end driving moved from single-trajectory regression [13, 15] to multimodal heads (anchored diffusion in DiffusionDrive [25]) and to *scoring* a fixed or generated candidate set. VADv2 scores a vocabulary of 4,096 trajectories with a learned probability [6]; Hydra-MDP distils rule-based metric teachers into a per-candidate scorer over an 8,192-entry vocabulary and reports that per-family sub-scores beat one scalar [22]; GTRS and DriveSuprim score dense vocabularies

coarse-to-fine [24, 42], and SparseDriveV2 composes 1,024 paths and 256 speed profiles into 262,144 trajectories scored coarse-then-compose [34]; WoTE scores candidates with a BEV world model [20]; ZTRS learns the scorer from reward alone [23]; GoalFlow conditions generation on a goal point [41]. The shared lesson is that separating *what can be proposed* from *what is chosen* pays, and that the gap between the two is large. In our own reference planner (an anchored-diffusion arm, REF-C, parity lineage, 40 validation episodes) the best candidate in a 256-anchor fan has an ADE@2 s of 0.1640 m while the candidate it selects has 0.4714 m[I] (registry §4; introduction), so the gap between proposing and choosing is larger than the gap between planners. None of these systems puts a tree over a versioned token vocabulary or chooses its sensors per tick (GTRS, DriveSuprim and SparseDriveV2 are coarse-to-fine).

Vision–language(–action) driving. EMMA [14], DriveVLM [35], SimLingo [33] and Alpamayo-R1 [28] decode trajectories or actions autoregressively from a VLM, often with reasoning traces; DriveVLM-Dual and FASIONAD pair a slow reasoner with a fast planner [31, 35]. Autoregressive decode is memory-bandwidth-bound on an edge SoC, which is the regime a vehicle computer lives in (§8). HiCAP keeps the VLM's world knowledge but replaces generation by a single prefill pass and a retrieval over cached action embeddings.

Contrastive and energy-based decision models. Contrastive Language Models (CLM) freeze a language model, train two small projection heads with bidirectional InfoNCE on cached embeddings, and choose the action whose embedding best matches the state; action embeddings are cached, and as a best-of-N verifier the model reaches 81.6% on DeepSWE and 87.6% on Terminal-Bench 2.1 at 4.1–5.7× lower latency than a generative verifier [18]. Its closest robotics relative is CoVer-VLA, a frozen SigLIP 2 with attention pooling over patch tokens, a numeric action-chunk tower and symmetric InfoNCE that verifies VLA action candidates [19]. Earlier, Implicit Behavioural Cloning used an InfoNCE energy over sampled counter-examples [10], later out-performed by diffusion policies [7], and VINN retrieves nearest-neighbour actions [29]. Contrastive RL interprets the critic as a discounted future-state occupancy [8, 38]. On the theory side the InfoNCE optimum is a pointwise mutual information [30, 36], the long-tail prior is removed by logit adjustment [26], and dot-product retrieval has provable dimensionality limits [39] that motivate late-interaction re-ranking [16]. Prior contrastive selectors treat the action set as flat, text or a small numeric chunk, and none imposes a prior over a hierarchical vocabulary or budgets its sensors (the Contrastive RL critic does predict outcomes in a shared space, but over goals, not over candidate actions). Trees of node scorers with beam retrieval are established in language modelling and retrieval, from the hierarchical softmax [27] to tree-based deep models and their beam-optimal training [44, 45].

Latent world models and imagination. Latent prediction on top of frozen or self-supervised encoders (V-JEPA 2 and its action-conditioned variant [2], DINO-WM [43], Dreamer 4 [12]) supports planning without pixel decoding, and world modelling as dense supervision amplifies data scaling for driving VLAs [21]. Anti-collapse without heuristics is provided by sketched isotropic regularisation [3]. Our own programme measured two failure modes that this literature does not foreground: a world model that faithfully rolls out an implausible candidate (consistency is not plausibility) and imagination confidence that rises as rollout fidelity decays. HiCAP therefore restricts imagination to what data can identify (Proposition 9): outcome-aware soft targets, an analytic cost from a forecast of a few non-reactive exogenous descriptors, and a gated critic with a disagreement veto.

Elastic sensing, token budgets, and audio. Token merging and pruning reduce visual tokens at small accuracy cost [4, 5]; Matryoshka representations make one embedding usable at several dimensions [17]. Multi-camera driving VLMs fix a camera set; we instead *choose* the set per tick. Audio enters language models either through a frozen speech encoder and projector [32] or natively in omni-modal models; joint audio–visual embedding spaces are studied in [11]. Driving-relevant sounds (sirens, horns) are absent from standard driving corpora, so the audio path must be built and controlled differently (§6.3).

Safety layers. Runtime assurance around a learned planner uses control barrier functions [40] and distribution-free coverage sets [1]; HiCAP's candidate set is a natural place to apply both before selection.

3 Problem setting and data interface

Decision problem. At each tick t the system receives a state $s_t = (I_t^{\mathcal{S}_t}, e_t, n_t, u_t)$: frames $I_t^{\mathcal{S}_t}$ from a chosen camera subset $\mathcal{S}_t$ (three frames, 100 ms apart, per camera); the ego kinematic state $e_t = (v, a_{\text{lon}}, \dot{\psi})$

at the cycle time; a navigation command $n_t \in \{\text{follow}, \text{left}, \text{right}\}$ with optional range and time arguments; and, in the audio arm only, a window of microphone samples u_t. It outputs a 6 s ego trajectory $\hat{y} \in \mathbb{R}^{8\times 2}$ at the slots $t_k \in \{0.5, 1, 1.5, 2, 3, 4, 5, 6\}$ s in the ego frame, which is the trajectory contract of the programme's planner line. The decision is a *leaf* $a = (v_1, v_2, v_3)$ of the action tree of §4.2 plus a small continuous refinement $\delta \in \mathbb{R}^{8\times 2}$: $\hat{y} = \pi_{\text{prior}}(e_t) + \rho(v_3) + \delta$, where π_{prior} is a constant-turn-rate-and-acceleration (CTRA) roll-out of the ego state and $\rho(v_3)$ is the residual profile of the leaf's code.

Data. We build on the newest programme corpus, not on the older parity corpus that the repository's default branch still describes. The corpus is the private Hugging Face dataset `tanitad-v7-training-corpus`: 4,719 clips, 26.2 hours of front-camera driving, 4,572 train and 147 evaluation clips, corpus id `a48251e89c7a8603`, label schema `s2-geom-v7`, released as v7, v7.1, v7.2 and v8.1 (the vocabulary was identical across the v7–v8.1 releases; the v9 release, §3.1, is unread)[I]. It is deliberately a *new comparability domain*: it does not re-select the canonical parity corpus of 2,376 episodes, and arms trained on it are compared only with arms trained on it. Frames are a cylindrical projection at 256×640 pixels covering a true 120° field of view (linear in azimuth, $f_{\text{ref}} = 305.577$), with a mandatory per-clip vertical crop because two camera rigs differ by about 215 pixels. Only the front-wide camera is cached; the upstream dataset has seven cameras, which matters for §6. None of the 36 upstream features is audio and no label or cache carries it; the audio stream inside the shipped video container is unprobed (UNVERIFIED).

What the labels are, and what they are not. Labels may use privileged quantities (ego future, other agents); inference may not: it sees frames, the ego state admitted by the PI on 2026-09-02, the stamped navigation command and, optionally, the audio channel admitted on 2026-09-29. Table 1 states, for every quantity, whether it is a training label, an admissible inference input, or an evaluation stratifier only. The strategic and tactical tokens are derived offline from the *ego's own future path* (geometry), so they are legitimate targets and inadmissible inputs. The perception-style tactical goals (traffic-light colours, gap targets, oncoming reactions) are teacher signals parsed from Alpamayo's chain-of-causation text; 49.6% of records carry at least one goal marked disputed and traffic-light colour was never independently verified[I]. They may supervise auxiliary heads but never enter as inputs, and every head on them carries an ego-only control.

Information-disjointness of the goal path. Two programme rules bind the design. First, a route or goal input may not carry the output of a situation classifier, otherwise attribution between goal and classifier is lost and the classifier is scored through the planner. HiCAP contains no situation-classifier *head*, and its only route-side input is the stamped navigation command; evaluation-time situation strata (`tac_SIT`) never re-enter. But the tactical cells and the 22-goal head share the readout $\mathbf{s}_2$ of one trunk, and fourteen of those goals are teacher-text situation-like labels (traffic-light colours, yield, gap, merge). We therefore do not claim disjointness: only geometry-derived goals may act on the decision (rule R4), the teacher-text goals are auxiliary with no path into any decision, and a goal-shuffled control and a strategic-token-null ablation bound what the goal path carries (NC6). Second, the parent's selection restricts the child candidates (a hard beam) but is not fed to the child's state readout, so the child level cannot echo the parent's decision as a feature; the control "flat tactical" (no parent restriction) tests whether the restriction itself matters (§10).

3.1 Training on the programme corpus

HiCAP is trained on the programme's own corpus with the newest tactical-label release, v9, which the PI names as the release behind refcv8 and v8f. We read its public-to-us files (private Hugging Face datasets `Sayood/tanitad-refcv8-v9-labels` and `Sayood/tanitad-v7-training-corpus`: cards, schema, validation report, manifests); the label arrays themselves are served from a file host the environment still blocks, so counts below are those the release's own manifests and report state (INHERITED from the release builders' MEASURED report, not re-verified here). The release covers 4,369 train and 139 held-out evaluation clips, 869,278 train rows of the trainer's 10 Hz clock (≈ 24.1 h), images and poses unchanged from the v7 corpus. It replaces the v8 labels, which the refcv7 audit found anchored at a single instant per clip: tactical labels existed on only 23.22% of training windows and the navigation input was wrong on 74.3% of left/right windows. v9 labels *every* frame over $[\text{NOW} + 2, \text{NOW} + 8]$ s: a lateral action (LANE_KEEP, TURN_L, TURN_R; lane change could not be measured and is never positive), a longitudinal action (HOLD, CREEP, STOP, FOLLOW, DECELERATE, ACCELERATE, KEEP) with target speed, time and distance, lead gap and time gap, the 22-token goal family per frame, a per-frame navigation command with distances, and a *route checkpoint* (the next route point in the vehicle frame, recommended RC-A50 with training noise $\sigma = 2$ m along, 0.75 m lateral, and dropout ≥ 0.3). Tactical labels cover 95.09% of train and 94.57% of evaluation frames; turn-side

Table 1: Admissibility of each quantity. *Label* = offline supervision (privileged signals allowed). *Input* = available at inference. *Strat.* = evaluation stratifier only. The navigation command is an oracle input by construction (it is the ego's own future turn), admitted under a programme ruling as a first-class route signal and stamped as such in every run; the nav-zero and nav-shuffled controls quantify what it buys.

quantity	source	role	note
frames (front-wide, 3×100 ms; further cameras when cached)	sensor	Input	cylindrical 256 × 640, per-clip crop
ego state e_t: $v, a_{\text{lon}}, \dot{\psi}$ at cycle time	sensor/CAN	Input	2026-09-02 ruling (velocity), extended by the PI (2026-10-25) to longitudinal acceleration and yaw rate; *curvature is excluded*; null row
navigation command n_t	ego-future	Input (stamped oracle)	controls: nav-zero, nav-shuffled
route checkpoint (v9 RC-A50, noised)	ego driven path	Input (stamped oracle)	PI decision 11 pending; controls: checkpoint-zero, shuffled; never in the evaluation of a vision-only claim
per-frame navigation token and arguments (v9)	ego-future turn table	Input (stamped oracle)	replaces the v8 single-instant nav; nav-zero, nav-shuffled
audio window u_t	microphone	Input, optional	admitted at inference by PI ruling 2026-09-29 as an *optional* channel; the model must run bit-identically without it (null row); no corpus carries audio
strategic tokens $g_{\text{str}}, a_{\text{str}}$	ego-future geometry	Label	band $[8, 30]$ s; four goals and two actions empty
tactical actions $a^{\text{lat}}_{\text{tac}}, a^{\text{lon}}_{\text{tac}}$	ego-future geometry	Label	band $[2, 6]$ s; factored, never flattened
tactical goals g_{tac} (22, multi-label)	geometry + Alpamayo text	Label (aux)	22 exclusive pairs; eight goals with $n < 30$
goal point g_{tac}.anchor	ego-future	Label	predicted, never supplied
3-D agent tracks	`obstacle.offline` join	Label (cost heads)	96.83% of train clips; eval-side coverage unstated
`tac_SIT`, `disputed`, `alpamayo.*`	Alpamayo text	Strat. only	never an input
`nav_30s`, `speed_max_input`	ego-future	Label / oracle	not built into any arm here (echo of the ego's realised speed)
situation-classifier head	—	**absent**	no such head; the goal heads share the trunk readout, so disjointness is bounded by control NC6, not assumed

recall/precision is 0.980/0.988 on train; five registered checks are reported failed by the builders (evaluation coverage 0.4 pp short, lane change never-positive, navigation distance, a 6-s navigation bar, 881 of 113,946 turn windows unlabelled). Three consequences for this design. (i) Measured on the v9 arrays (md5-verified; `hicap_v9_screen_census.py`)[M]: the lateral family has three usable classes (lane keep 632,971 rows, turn left 48,830, turn right 47,726; lane-change entries exist in the vocabulary but are never positive) and the longitudinal family seven (hold, creep, stop, follow, decelerate, accelerate, keep; from 8,658 to 191,225 rows), so the tactical grid is $3 \times 7 = 21$ cells of which 19 are populated, all 19 with at least 30 train rows and 15 of 19 on the 139 evaluation clips; the 35-cell grid of the v7 vocabulary no longer applies. Label rows are 83.9% (lateral) and 71.3% (longitudinal) of train rows with a definite class; the remainder are ignore or partial labels (the release's 95.09% counts any tactical label). The tactical band is $[2, 8]$ s, not the $[2, 6]$ s of the v7 vocabulary used in Sections 4.2–9; the class set is the nine-class v9 family (with the frozen v7 ids supplied beside it), so the cell grid, the conformal keep-set calibration and every count in Appendix A are re-derived from the v9 arrays before H-HC0; they are v7 figures until then. (ii) Two new inputs are oracle by construction and are stamped like the navigation command: the per-frame navigation token with its arguments and the route checkpoint, each with a zeroed and a shuffled control; the builders' own study found a planner that merely aims at the checkpoint turns the right way on 0.89–0.97 of turn windows (refcv7: 0.841), so "beats refcv7" proves nothing without the checkpoint-zero row, and whether road geometry is admissible inside the route input (their decision 11) and whether eight recording-partner clips are dropped (decision 8, default keep) are still pending. (iii) The strategic level (goals and actions over 8–30 s) is not part of the v9 release and keeps coming from the v7/v8.1 corpus release. The training plan has four steps, each with a stated cost. (1) Build the backbone cache once per backbone and camera set (§8); this is the only step that needs the large backbone. (2) Gate H-HC0 on the cache. (3) Train the ≈ 27 M trainable part on cached tokens through the three-stage curriculum with the level-wide losses of §4.3: stage 1 on the strategic and tactical levels with soft targets, stage 2 adding the factored trajectory levels and cost heads, stage 3 hard negatives and the refinement head with 40% replay; one run of the 56 M trainable part is GPU-minutes to GPU-hours on cached tensors, not the ≈ 45 GPU-hours of the flat reference. (4) Evaluate against refcv8

and v8f on the identical windows of the v9 evaluation split, all four metric families, paired episode-cluster bootstrap on the full-set mean, each arm with a seed-1 replicate; the v8f arms are the hierarchy-attention reference. Evaluation clips are never used for any fit, and the contamination screen of §8.6 runs on them first.

4 Method

Figure 1 shows the system. A frozen VLM turns camera frames into semantic tokens at a slow rate; a tiny trainable motion adapter supplies what a frozen single-frame model cannot; a small fusion transformer with level queries produces one unit-norm state readout per decision level; and the action side is a tree of cached embeddings searched top-down under a chain of prior-encoded pruning rules. Language never generates anything: the language tower embeds token definitions once, and is used at run time only by the optional System-2 verifier (§4.4).

4.1 Backbone, motion adapter and fusion

Frozen semantic backbone. The backbone $\mathcal{V}$ is a frozen vision–language model. At the slow rate it reads the selected cameras and a fixed short prompt once (prefill only) and we cache two things: the hidden states of the visual tokens at three depths of the language model (middle, three quarters, final), pooled to at most $N_v = 128$ tokens by fixed adaptive pooling, and the last-token hidden state (the CLM readout). Recent frozen-backbone work favours a learned aggregator over bare last-token pooling [19] and reading several depths, so the fusion transformer reads the token sets, not one vector. The PI has ruled that a VLM is the backbone and that its size is bounded by latency, not by a parameter count; which member of the Cosmos family and its Qwen3-VL relatives is used is decided by measured latency and paired ablations (§8.6, §8.3), and the interface is fixed regardless of the choice (visual tokens of width d_v, a text tower, optionally an audio input).

Motion adapter. Frozen image and video encoders were measured to fail at metric ego-motion in this programme (a frozen single-frame encoder scored 2.1675 m ADE, REF-A[I]), and the ego block is admitted by the programme's 2026-09-02 ruling that velocity at cycle time is a legal initial state (extended by the PI on 2026-10-25 to longitudinal acceleration and yaw rate; curvature is excluded and is not fed, §12); the adapter exists to recover metric motion from frames, measured by H-HC1 with the ego block zeroed, and the ego-zeroed row is reported for every family. The adapter M_ϕ (≈ 2 M parameters) therefore takes the three-frame stack (100 ms apart) at 64×160, with inter-frame differences, and returns $N_m = 16$ motion tokens. It is the only vision component that is trained, and it is trained on frame pairs, not on single frames.

Fusion transformer. F_θ ($d = 512$, six layers, eight heads, ≈ 19 M parameters[E]) reads: the N_v projected VLM tokens (each with a Fourier embedding of its *age*, since VLM tokens are reused between refreshes), the N_m motion tokens, one ego token (an MLP over e_t with a learned null row for ego-dropout), one navigation token, and learned level queries. Audio is not among the input tokens: the level-query outputs receive it through one zero-initialised gated cross-attention read (§6.3), so that the model without audio is exactly the original. The state readouts are

$$\mathbf{s}_\ell = \mathrm{norm}\left(H_\ell(F_\theta(\cdot)[q_\ell])\right), \qquad H_\ell : \mathbb{R}^{512} \to \mathbb{R}^{1024} \to \mathbb{R}^{512}, \tag{1}$$

with the tactical level split into a lateral and a longitudinal query $(\mathbf{s}_2^{\mathrm{lat}}, \mathbf{s}_2^{\mathrm{lon}})$ and the trajectory level into a path and a profile query $(\mathbf{s}_3^P, \mathbf{s}_3^V)$, because the programme measured that a single softmax mixing lateral and longitudinal destroys the longitudinal decision. Readouts are Matryoshka-nested at widths $\{128, 256, 512\}$ as a free option (§6). The trainable part of the model is 56.4 M parameters, counted from the implementation at the default widths[M] (`stack/tanitad/hicap`): fusion transformer 18.9 M, cached-token projector 4.6 M, motion adapter 0.6 M, ego, navigation and route tokens 0.9 M, optional gated audio read 1.4 M, five readout heads 5.3 M, three text-node towers 8.1 M, numeric path and profile towers 1.1 M, goal, cost, refinement and pair heads 1.9 M, candidate re-ranker 13.7 M. The earlier estimate of 27 M omitted the projector, the towers and the re-ranker; the re-ranker is trained only in stage 3, so stages 1–2 train 42.7 M. Readout rates follow the horizons they serve: strategic every 2 s, tactical every 0.5 s (with the VLM refresh), trajectory every 0.1 s.

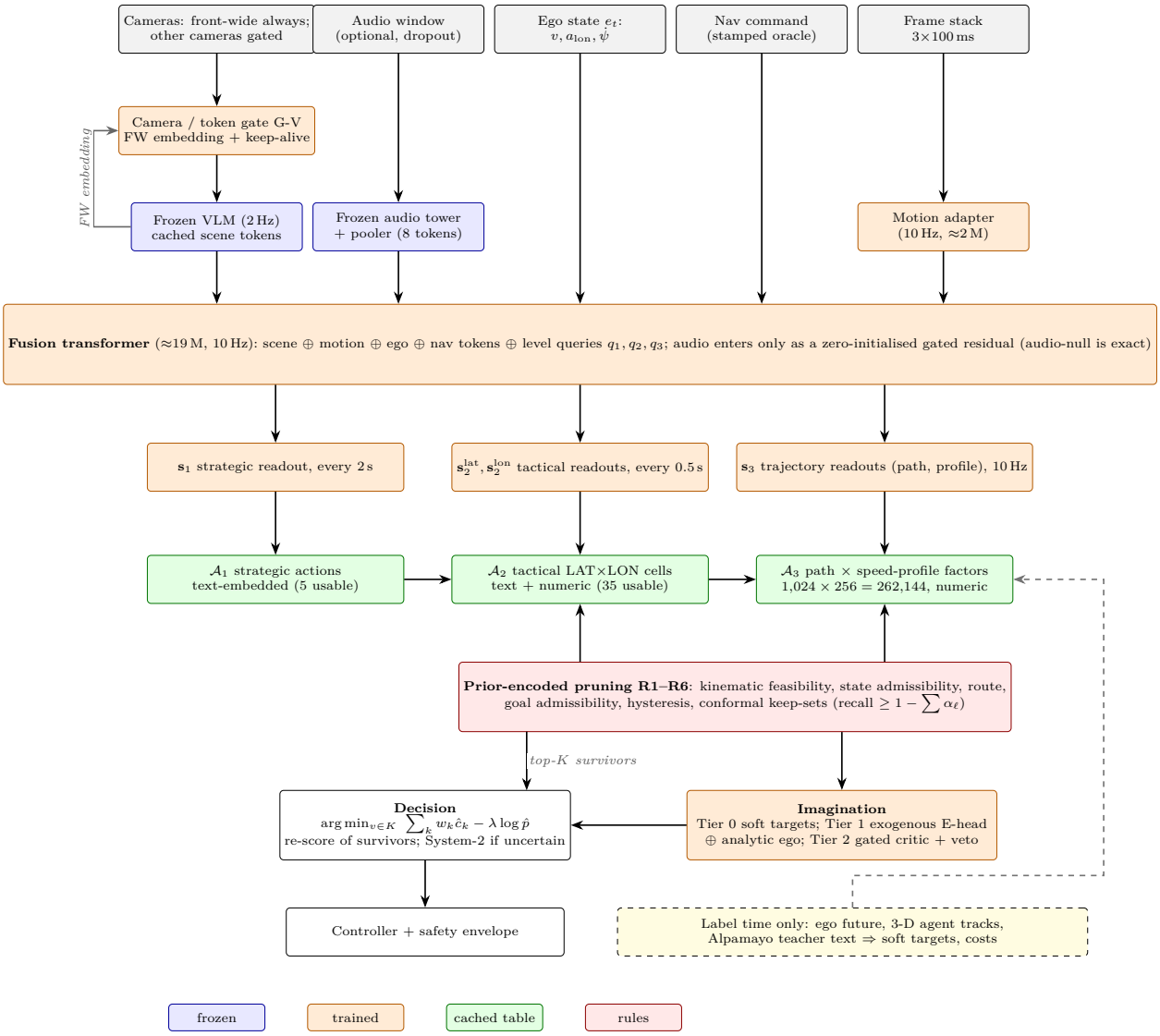


Figure 1: HiCAP. Blue: frozen. Orange: trained. Green: cached embedding tables of the action tree. Pink: pruning and admissibility rules. The VLM refreshes at 2 Hz, the motion adapter and fusion at 10 Hz, and the level readouts update at 0.5/2/10 Hz. A parent's choice restricts its children (a hard beam) but is not an input to the child's readout. Survivors of the pruning cascade are returned to the fusion transformer as candidate tokens for re-scoring; a System-2 verifier runs only when the top-two margin is small or the two imagination heads disagree. The dashed box is label-time supervision, never available at inference.

Table 2: The action tree. *Declared/populated* counts are per-field over the 4,719 clips of the v7 corpus[I]; a token with $n = 0$ is masked in the loss and in retrieval (head width stays full). Level 3 is a design vocabulary, not a shipped one; its sizes are proposals fixed in a pre-registration amendment.

level	declared / populated	source	band, rate	node embedding
L1 strategic action a_{str}	7 / 5	ego-future geometry (label)	$[8, 30]$ s, every 2 s	frozen text tower on the token definition $\oplus$ arguments (`within_m`, `by_time_s`) $\rightarrow$ MLP
L2 tactical cell $(a^{\text{lat}}_{\text{tac}}, a^{\text{lon}}_{\text{tac}})$	$8 \times 8 = 64$ / $5 \times 7 = 35$	ego-future geometry (label)	$[2, 6]$ s, every 0.5 s	two factored embeddings (lat, lon) from text $\oplus$ arguments (`v_target_ms`, `within_m`); pair-admissibility bias $b(c_{\text{lat}}, c_{\text{lon}})$
(side) tactical goals g_{tac}	22 / 22 (eight with $n < 30$)	geometry + Alpamayo text (aux label)	$[2, 6]$ s	multi-label head on $\mathbf{s}_2$; 22 exclusive pairs; goal point (x, y, t_{reach}) regressed
L3 path factor $\mathcal{P}$	K_P = 1,024, partitioned by lateral class	train paths on an arc-length grid	$[0, 6]$ s, every 0.1 s	numeric MLP on the path's curvature profile
L3 profile factor $\mathcal{V}$	$K_V = 256$, partitioned by longitudinal class	train speed-ratio profiles over the CTRA prior	same	numeric MLP on the profile; pair head for (p, q)

4.2 The action tree and its vocabularies

The PI's instruction was to use the last tactical vocabulary of the programme's dataset as the action space. That vocabulary is the frozen v7 vocabulary (`vocab_v7.py`, frozen 2026-08-23): 56 declared slots ($8 + 7 + 22 + 8 + 8 + 3$), 52 distinct strings, ids equal to tuple indices, extension only through a new versioned tuple[I]. Table 2 maps it onto the tree; the full token lists with counts are in Appendix A.

Why these levels, and their limits. The operative band $[0, 2]$ s has no tokens by design in the programme's vocabulary; it is a continuous quantity, which is what Level 3 encodes. The strategic supply is thin: four of eight strategic goals and two of seven strategic actions have zero examples (no map, no time expression in the source text), leaving five usable strategic actions with `HOLD_MAIN_ROAD` alone at 2,468 of 4,719 clips[I]; lane changes are unreachable as tactical lateral actions because the emitter has no branch for them. So Level 1 is a five-way decision today. Strategic labels are also truncated by clip length: clips last about 20 s with the anchor at 8 s, so at most about 12 s of future exists and Level 1 is effectively an $[8, 12]$ s decision on this corpus, reported with its per-window n. We keep the level because it is the programme's thesis and the tree admits growth without redesign, but every strategic claim in this paper is conditional on that supply, and the pre-registration reports the strategic family with its n and reason (§10). *Band mismatch.* The tactical tokens describe the $[2, 6]$ s band, while a candidate is a 6 s object. We resolve it by minting each candidate's tactical cell over $[2, 6]$ s with the label emitter's thresholds, and the leaf refines $[0, 2]$ s; the agreement of these minted cells with the shipped labels on the validation windows is a pre-registered check, not an assumption.

Node embeddings. For Levels 1 and 2 the embedding is $\mathbf{a}_v = \text{norm}\left(G_\ell([\mathbf{t}_v; \varphi(\text{args}_v)])\right)$, where $\mathbf{t}_v$ is the last-token hidden state of the frozen language tower on the token's definition string (computed once and cached) and φ a Fourier feature map of the numeric arguments. Text embedding gives structured geometry for free (`PREPARE_TURN_L` lands near `TURN_L`) and lets a new versioned token be added with a meaningful initialisation. The tactical cell is scored *factored*,

$$f_2\big(s, (c_{\text{lat}}, c_{\text{lon}})\big) = \tau_2 \langle \mathbf{s}_2^{\text{lat}}, \mathbf{a}_{c_{\text{lat}}} \rangle + \tau_2 \langle \mathbf{s}_2^{\text{lon}}, \mathbf{a}_{c_{\text{lon}}} \rangle + b(c_{\text{lat}}, c_{\text{lon}}), \tag{2}$$

with $b = -\infty$ for pairs that a joint lateral-by-longitudinal census shows never occur (the programme's goal-admissibility tables are goal-to-action maps, not pair tables, and $5 \times 7 = 35$ is a product of marginally populated classes) and a learned scalar elsewhere, so the lateral and longitudinal decisions remain separately evaluable as the marginals of one distribution.

The trajectory vocabulary is a factored product. A trajectory is composed from a *path* $p \in \mathcal{P}$ (geometry on an arc-length grid, $K_P = 1{,}024$ entries) and a *speed profile* $q \in \mathcal{V}$ (the speed change $v(t) - v_0$

over the horizon, a residual over constant speed, $K_V = 256$ entries, with a dedicated standstill sub-vocabulary because a ratio $v(t)/v_0$ is singular at $v_0 \to 0$: hold, creep and stop) by arc-length re-parameterisation, $N = K_P K_V = 262{,}144$ composed entries. This is the structure and size of the densest planning vocabulary we found in the literature (search scope stated in §11), scored coarse-then-compose [34], and it aligns with the v7 tokens: the lateral tactical class partitions $\mathcal{P}$ and the longitudinal class partitions $\mathcal{V}$, so Level 2 chooses which path and profile subsets are live. Lateral and longitudinal structure are exactly what the tactical tokens factor, and 88.7% of the programme's oracle gap is longitudinal[I], so the profile factor is where vocabulary capacity should be spent. Both factors are built from *training-split* windows by farthest-point sampling *within class*, with a floor per class, because k-means starves rare classes; the factors are labelled by the same geometric classifier that mints the training labels, so that the index and the labels cannot disagree. The speed-relative profiles are what make the physical prior cheap (a 3.4× smaller oracle-in-set error at 256 entries in §9). A small fine stage models what a product cannot, the coupling of the two factors (lateral acceleration $v^2\kappa$, braking before a curve), and a residual refinement head on the chosen entry escapes quantisation. Growth beyond 262,144 (for example $2{,}048 \times 512 \approx 1.05\,\mathrm{M}$) is gated on the oracle-in-set curve still falling (H-HC5). The earlier alternative, a per-cell two-stage residual code, is kept as an ablation. The shipped labels cover only $\pm 2\,\mathrm{s}$ around each clip's anchor; for other windows the cell is recomputed from the window's own future with the emitter's threshold rules (label-side only), which is an integration item because the emitter lives on a branch not yet in the repository (§12).

4.3 Scoring and training

Level-wide critics. Beam retrieval ranks nodes *across* parents, so each level needs a critic that is a *marginal*, and two traps follow. (a) The prior added at selection time must be the marginal prior $\bar{q}_\ell(v)$ of the node (the mass of its subtree), not the conditional $\hat{q}_\ell(v \mid u)$. (b) A softmax over siblings is invariant to any per-(state, parent) offset, so training with sibling-only negatives leaves that offset free and mis-orders nodes across parents; each level is therefore trained with *level-wide* negatives (a softmax over all nodes of the level). With a dot-product critic a mean-pooled centroid cannot serve as a node embedding, since it scores the *average* child and hides a narrow peak inside a large cell; node embeddings are learned as marginals (Proposition 4, and the toy of §9). For level ℓ,

$$\tilde{p}_\ell(v \mid s) = \mathrm{softmax}_{v' \in V_\ell} \Big(\tau_\ell \langle \mathbf{s}_\ell, \mathbf{a}_{v'} \rangle + \beta_\ell \log \bar{q}_\ell(v') \Big)_v, \tag{3}$$

with $\tau_\ell = e^{\theta_\ell}$ initialised at $1/0.07$ and clamped at 100 as in CLM, and $\bar{q}_\ell$ the train-split marginal frequency. By Corollary 2 the InfoNCE-trained critic carries $-\log \bar{p}$, so β_ℓ is fitted on held-out training episodes (grid $\{0, 0.25, 0.5, 1, 2\}$); with the soft-target softmax loss below the prior is already absorbed and $\beta_\ell = 0$ is the default. With roughly 1.5 training windows per leaf the prior is estimated hierarchically (a Dirichlet-tree backoff, $\hat{q}(a) = \hat{q}(S)\,\hat{q}(\mathrm{cell} \mid S)\,[(c_{\mathrm{leaf}} + \kappa/L_f)/(c_{\mathrm{cell}} + \kappa)]$ with $\kappa \approx 0.5$), not from a flat smoothed histogram. The conditional cascade distribution $\hat{p}_\ell(v \mid u, s)$, the softmax over the kept children of the logged parent, is tied to the level-wide critic by the consistency loss below.

Losses. The total loss is

$$\mathcal{L} = \sum_{\ell=1}^{3} \lambda_\ell \mathcal{L}_\ell^{\mathrm{voc}} + \lambda_{\mathrm{c}} \mathcal{L}^{\mathrm{cons}} + \lambda_{\mathrm{g}} \mathcal{L}^{\mathrm{goal}} + \lambda_{\mathrm{k}} \mathcal{L}^{\mathrm{cost}} + \lambda_{\mathrm{r}} \mathcal{L}^{\mathrm{ref}} + \lambda_{\mathrm{i}} \mathcal{L}^{\mathrm{img}} + \lambda_{\mathrm{a}} \mathcal{L}^{\mathrm{aud}} + \lambda_{\mathrm{f}} \mathcal{L}^{\mathrm{cfg}}, \tag{4}$$

where $\mathcal{L}^{\mathrm{img}}$, $\mathcal{L}^{\mathrm{aud}}$ and $\mathcal{L}^{\mathrm{cfg}}$ are defined in §5 and §6.

- **Vocabulary loss.** $\mathcal{L}_\ell^{\mathrm{voc}} = -\sum_{v \in V_\ell} y_\ell(v) \log \tilde{p}_\ell(v \mid s)$ with level-wise soft targets $y_\ell(v) = \sum_{a \in \mathrm{subtree}(v)} y(a)$, where $y(a) \propto \exp\big(-d_{\mathrm{tw}}(y^\star, \hat{y}_a)^2 / 2\sigma^2\big)$ is the distance-weighted target of VADv2 [6] with $\sigma = 0.5\,\mathrm{m}$, extended upward to ancestors by summation. Levels 1–2 use the label, masked for empirically absent classes. Any node whose subtree holds a candidate within 0.25 m of the logged path receives target mass through y and is *not* removed from the normaliser, since removal would change the optimum (Remark after Proposition 1); false negatives are most frequent at coarse levels. The physical-tier masks of §4.4 are applied to the logits in training as at inference, so that $p(a \mid s, \mathrm{feasible})$ is what is learned.
- **Contrastive stage-1 variant.** In stage 1 the leaf loss can be the bidirectional in-batch InfoNCE of CLM with the same false-negative mask; Without the mask Corollary 2 says what it estimates; with the mask the optimum changes (Remark after Proposition 1) and β must be fitted, not set to one. Masked InfoNCE, unmasked InfoNCE and the soft-target softmax are registered as an ablation (H-HC4), not assumed.
- **Goal loss.** A multi-label head on $\mathbf{s}_2$ over the 22 tactical goals, binary cross-entropy with positive weight capped at 50, negatives supervised only for goals emitted from geometry or entailed false by an exclusive

pair, plus an exclusivity penalty; and a regression of the goal point at 2/4/6 s. Goals are *predicted*, never supplied.

- **Consistency.** $\mathcal{L}^{\text{cons}} = \sum_\ell \mathrm{KL}\big(\,\mathrm{sg}[\hat{p}_\ell^{\text{cascade}}]\,\|\,\tilde{p}_\ell\big)$ ties the level-wide critics to the cascade marginal.
- **Cost heads (Hydra-style).** Per composed entry, heads $\hat{c}_k(s,a) = \mathrm{MLP}_k(\mathbf{s}_3 \odot \mathbf{a}_a)$ predict privileged costs: minimum time-to-collision proxy and minimum clearance, computed from the logged *agent tracks* replayed non-reactively against each entry's path (§5), and progress deficit. Comfort (acceleration and jerk) is computed exactly from $\hat{y}$ and needs no head. Distilling per-family sub-scores rather than one scalar follows Li et al. [22].
- **Refinement.** $\delta = g(\mathbf{s}_3, \mathbf{a}_a)$, L_1 to the residual left after the entry. It turns the vocabulary into a set of *open* candidates, so the vocabulary bounds the search, not the accuracy.

Curriculum. Three stages on cached features, as in CLM. Stage 1 trains all levels with level-wide negatives and no hard negatives. Stage 2 adds mined near-miss negatives (entries whose ADE against the logged path lies in $[0.25, 1.0]$ m, plus synthetic speed-scaled copies of the positive) with 40% stage-1 replay; CLM reports that hard negatives from the start overfit (62.4% against 69.2%)[P] (release README, not re-run here). Stage 3 unfreezes fusion and heads jointly and adds cost, refinement, imagination and audio losses, again with replay. Optimisation follows the programme's pre-registration: AdamW, no weight decay, lr $= 2\times10^{-3}\sqrt{1024/\text{width}}\sqrt{\text{batch}/1024}$, one-cycle with 10% warm-up, batch 2,048, three training seeds per arm plus one further replicate seed for the noise floor; the headline is the mean of the three, with the interval pooled over episodes, never over seeds.

Decision rule.

$$a^\star = \arg\min_{a\in K} \sum_k w_k\, \hat{c}_k(s,a) \;-\; \lambda \log \hat{p}(a \mid s), \qquad \hat{p}(a \mid s) = \prod_\ell \hat{p}_\ell(v_\ell \mid v_{\ell-1}, s), \tag{5}$$

with the cost weights w_k fixed *a priori* by the four-metric-family policy and λ fitted on held-out training episodes, never on validation data. With $w = 0$ this is the maximum-likelihood entry among those retained by the beam.

4.4 Prior-encoded pruning and candidate-conditioned re-scoring

The candidate space is reduced by a short, ordered chain of rules, each stated so that its failure mode is measurable. Rules R1–R3 are hard and cheap; R4–R5 are soft; R6 is a calibrated guarantee. All act as *bitsets* over the two factors (a 1,024-bit path mask and a 256-bit profile mask, ANDed), built once per tactical update from v_0, the initial acceleration and the yaw rate, and an analytic box test lets a mask act on a whole subtree without touching its leaves (in the toy it never removed a node containing a feasible leaf).

R1 Kinematic feasibility (hard = physical limits; comfort = soft cost).
Keep entries whose implied longitudinal acceleration, braking and lateral acceleration, given v_0, lie inside physical limits $\mathcal{B}_{\text{phys}} = (6, 10, 9)\,\text{m/s}^2$; add initial-acceleration and initial-curvature continuity tubes around the measured state. Comfort limits are exceeded by experts, so they enter the re-scorer as a soft penalty. Measured on real windows in §9: the physical mask removes 78% of a speed-agnostic anchor set and keeps every logged future.

R2 State-conditioned admissibility (hard, calibrated).
Keep tactical cells that the training data make plausible for the binned ego state, or, as a retrieval prior, the union of the cells of the expert actions of the k nearest cached training states; calibrate the threshold conformally (R6) so the logged cell is retained with a stated probability. The retrieval form needs no situation classifier: it conditions on the state itself.

R3 Route consistency (hard in the nav-fed arm only).
With $n_t =$ left, exclude right-turn cells and conversely. The navigation command is oracle-perfect in the corpus, so a hard rule looks better here than it will deploy; the nav-zero arm is the deployment-honest row and the rule is off there.

R4 Goal admissibility (soft).
R4 uses only the *geometry-derived* goals (follow lane, turn left or right, stop point, speed band): the predicted set contributes a soft log-prior bonus to the cells that serve those goals through the programme's goal-to-action tables, soft because a wrong predicted goal must not delete the right cell. The teacher-text goals (traffic-light colours, yield, gap, merge, oncoming reaction) are auxiliary heads with *no path into any decision*. The goal and tactical heads share the trunk readout $\mathbf{s}_2$, so a goal-shuffled control bounds what the goal path carries. Scene-type priors that would consume a situation classifier's output are not used.

R5 Hysteresis (soft).
Entries within a tube around the previous plan shifted one tick receive a prior bonus, and switching requires a margin, to suppress oscillation.

R6 Conformal keep-sets (guarantee).
Per level, keep the smallest set whose score threshold has calibrated coverage $1 - \alpha_\ell$ (Proposition 7), giving joint recall $\geq 1 - \sum_\ell \alpha_\ell$. Calibration is on *episode blocks* and *class-conditional* (Mondrian): a marginal guarantee hides the rare classes, for which the toy showed lower survival (stop and turn states).

The width of each beam is not fixed but follows the calibrated margin of the level's critic (*compute follows uncertainty*). A small *guard set* (hold-current CTRA, gentle brake, firm brake, maximum brake in lane, the previous plan shifted one tick) is injected after every mask, so the candidate set is never empty and the recall floor is a safe fallback rather than zero.

Cascade and cadence. S0 ego masks (a bitset AND, every tick) → S1 factored coarse scoring every 0.5 s: classes first (5 lateral, 7 longitudinal), then members, keeping (k_P, k_V) of about $(20, 10)$ → S2 compose ≈ 200 entries with the pair mask and add the guard set; the survivors are cached for five ticks and re-masked each tick with the new v_0 → S3 late-interaction (MaxSim) shortlist to ≤ 32 → S4 cross-attention re-rank of K candidates. We train at $K = 32$ and deploy $K = 16$, escalating to 32 when the S1 margin is small. The strategic gate refreshes every 2 s. Cost per tick at $K = 16$ is ≈ 0.34 GMAC ([E], §8).

Feeding the survivors back to the backbone. The pruned set is not only scored by the dot product. Candidate tokens, each the cached action embedding, the cell embedding and kinematic descriptors, attend to the scene tokens in a three-layer light decoder (queries are candidates, keys and values are the fusion transformer's tokens), with a mixture-of-experts feed-forward layer routed *deterministically* by the candidate's tactical class and per-candidate sub-score heads; late interaction over a small set is the standard remedy for the rank cap of a single dot product [16, 39], and the cost is dominated by a scene projection that does not depend on K. Optionally, a *System-2 verifier* runs when the top-two margin is below $m_{\min}$ or the imagination heads disagree: the top $K_2 \leq 8$ survivors are rendered as short text (*lat=LANE_KEEP, lon=BRAKE_TO, target 6 m/s*) or as tactical-cell tokens appended after the cached scene prefix, one extra weight pass of the VLM reads the scene and each candidate, and the CLM match score is added to the re-score. The verifier fires only in uncertain ticks and only at tactical cadence or slower; it is the only place where the language side of the backbone acts at inference. A weight pass through an 8 B model costs ≈ 58.6 ms in bf16 and ≈ 14.7 ms at 4 bits at the Thor's bandwidth ([E]).

5 Imagination that can be trusted

Prediction structure and imagination are the natural way to give a selector a sense of consequence: rather than asking "which candidate resembles what the expert did", ask "what happens if I do this". The obstacle is identification. A world model trained on expert logs sees each action only in the states where the expert chose it, so its answer for any other action is an extrapolation. Our programme measured two consequences of trusting such extrapolations: a rollout can be perfectly self-consistent for a candidate that is implausible, and rollout confidence can rise while rollout fidelity falls[I]. The imagination stream of this design (repository manifest in §9) added a third, uncomfortable finding that shapes the whole section: in a controlled toy, *most of what looks like the benefit of imagination is the shape of the training target*, which costs nothing at inference. HiCAP-I is therefore built in three tiers, cheapest and safest first.

Tier 0 (training only): outcome-aware soft targets. The expert's one-hot label is the worst target the programme has measured for a selector[I]. Tier 0 replaces it by a listwise target built from per-candidate outcomes, which labels may use freely (ego and other agents),

$$y(a) \;=\; 0.7\,\frac{\exp(-J_{\text{teacher}}(a)/T)}{\sum_{a'}\exp(-J_{\text{teacher}}(a')/T)} \;+\; 0.3\,\mathbb{1}[a = a^\star], \tag{6}$$

where J_{teacher} is a cost computed from the logged future of other agents and the candidate's known kinematics, with one head per metric family (safety, comfort, progress, lateral). It has zero inference cost. In the toy of §9 it cut regret from 7.00 to 3.43 (paired difference 3.57, 95% interval $[2.99, 4.19]$), more than any imagination arm added on top.

Tier 1 (every tick): imagine the exogenous part, integrate the ego part analytically. For a candidate a the outcome is (W, e_a): the future scene W and the ego path $e_a = \kappa(a, e_t)$, a *known* function of the entry and the ego state. Only W needs to be learned, and under the non-reactive assumption of Proposition 9 it does not depend on a, so it is identified by logs alone. The exogenous part is represented by a handful of descriptors, not by embeddings or frames (a future frame contains the ego's own motion and would smuggle the action into the target): an E-head $\mathbb{R}^{512} \to \mathbb{R}^{1024} \to \mathbb{R}^{24}$ reading the state only outputs the lead gap, closing speed and lead acceleration at $0.5, 1, 2, 3$ s, a posterior over three to five behaviour modes (for instance the lead brakes, holds, accelerates) and onset-time quantiles. The cost of a candidate is the *expectation over the mode and the timing*,

$$J_1(s, a) = \mathbb{E}_{m,\tau \sim \hat{p}(\cdot|s)} \big[c\big(e_a,\ x_{m,\tau}\big) \big], \tag{7}$$

with c an analytic time-to-collision and clearance cost evaluated on the unicycle integration of the candidate. Marginalising *every* hidden factor matters: a sharp mode posterior with a mean timing was over-confident in the toy (+0.7 regret). Tier 1 needs logs only, no counterfactual data, and it is the one imagination arm that survives their absence (regret 1.80 against 47–72 for the unstructured arms). Its cost is ≈ 3 MFLOP per tick on the pruned set.

Tier 2 (gated, top-8, only where counterfactual supervision exists). Two cost heads on $\phi(s, a) = \mathrm{norm}(\mathbf{s} + R\,\mathbf{v}(a))$ regress per-family *counterfactual* costs of a candidate, obtained from a rule teacher or a simulator (Proposition 10 shows why the score cannot be an outcome-InfoNCE: it is identified only up to a per-candidate constant and cannot rank candidates, and a candidate-normalised softmax needs counterfactual data). Outcome InfoNCE is kept only as auxiliary representation shaping, and its targets are the exogenous descriptors of Tier 1 in the current ego frame, *never* embeddings of future frames (which contain the ego's own motion and would teach an echo); an action-shuffled control whose failure voids Tier 2 checks this. Tier 2 runs only for candidate families with counterfactual supervision, only on the survivors, and only when a *learned* benefit gate fires (target $\leq 30\%$ of ticks). The disagreement of its two heads is a *veto* on out-of-vocabulary candidates, not a benefit trigger: it detected them with an AUROC of 0.88 (recall 64% at 5% false positives), while need-times-trust and margin gates were no better than random as triggers, and a learned gate reached 65% of the gain at 30% of the budget.

Composition and losses.

$$\mathrm{score}(a) = \tau \langle \mathbf{u}(s), \mathbf{v}(a) \rangle + \log \mathrm{priors} - \beta_1 J_1(s, a) - \beta_2 J_2(s, a), \qquad J_1, J_2 \geq 0, \tag{8}$$

so each imagination term adds a non-negative cost to a candidate (which can still change the argmin, since a candidate the base score ranked lower may be penalised less), and each term carries a zero-weight ablation. The imagination losses are $\mathcal{L}^{\mathrm{img}} = \mathcal{L}^{\mathrm{exo}} + \mathcal{L}^{\mathrm{out}} + \lambda_{\mathrm{inv}} \mathcal{L}^{\mathrm{inv}} + \mathcal{L}^{\mathrm{cons}}$: a Huber loss on descriptors plus cross-entropy on the mode plus a quantile loss on onset; an auxiliary outcome InfoNCE against the exogenous descriptors at $k \in \{5, 10, 20, 40\}$ ticks, with per-state negatives where counterfactuals exist; an inverse-dynamics term of weight at most 0.1 that keeps the action recoverable from state and future; and a parent–child hinge that keeps child outcomes consistent with their parent (strategic outcomes cannot be labelled without a map, so no strategic outcome quality is claimed). *No gradient flows from any imagination head into the frozen backbone*, and imagination parameters are trained separately from the selector's, since a world model that scores its own planner's fan was measured to help by only +0.21[1].

Guards, each a measured failure. (G1) An analytic feasibility and comfort mask is applied *after* every learned term: imagination alone picked an infeasible over-braking candidate in 89.8% of toy ticks and the mask brought it to 0%. (G2) A counterfactual-coverage audit: a family without counterfactual supervision is refused by Tier 2. (G3) A shuffled-cue and shuffled-v_0 control, with R^2 reported on the *exogenous residual* only, because predicting the lead's speed from the ego state alone looks like imagination (echo). (G4) Disagreement is a veto, never a benefit gate. (G5) A confidence-sign audit: adding an imagined-cost term flipped the correlation of confidence with regret from -0.11 to $+0.19$ in all three seeds, so the combined softmax is never exposed as confidence. (G6) β is selected on held-out *training* episodes that carry counterfactual labels (never on validation data), or it silently returns zero. (G7) Every hidden factor is marginalised. (G8) Every term is reported per family with a paired episode-cluster interval and its n (agent-track coverage is stated for training clips only), and a risk-sensitive floor and a model-free value estimate on the same rows are always shown, otherwise risk-sensitivity is credited to imagination. (G9) The goal path stays information-disjoint from the situation classifier: the E-head reads the state only.

Cost, and the first experiment. The imagination overhead is ≈ 38 MFLOP per tick amortised (Tier 2 dominates), about 0.005% of the smallest prefill ([E]); calling a learned world model for every candidate, or iterating rollouts over the whole set, would be 3–10% of prefill and is not admitted. Efficiency is not the constraint; identifiability is. The decisive first experiment is a probe costing ≈ 0.1 GPU-hours on cached embeddings (E-I0): ridge and two-layer probes from the state to the lead gap now, the closing speed, the gap at +2 s and the time to brake on the validation windows, against a v_0-only baseline and a shuffled-cue control. If the frozen state does not carry metric gap and closing speed (the programme's earlier frozen encoders did not), Tier 1 is inert and only Tier 0 ships; the toy's whole gap between 3.00 and 0.46 regret is exactly this readout error.

What could go wrong, and how the protocol sees it. (i) Tier 1 may be inert because the backbone features lack range (E-I0, H-HC8). (ii) The programme's earlier attempt to train an agent detector jointly with the trunk failed a two-seed capacity gate at 17 M parameters and the deranged-join control reproduced the whole degradation[I]; here the heads sit on a frozen backbone and do not compete with a trunk, but that is a hypothesis. (iii) The non-reactive assumption fails at exactly the interactions that matter (yielding, merging); the one-sided form limits the damage and the shuffled controls (NC7) measure it. (iv) In the toy the factorised arm *lost* to the risk-sensitive floor when the cue was clean (regret 3.00 against 2.17 at a signal-to-noise ratio of 3) because of visible-state readout error, the same failure as an earlier programme stream.

6 Elastic sensing and audio

Not every camera, token or channel is worth its cost at every instant. Table 5 shows why it matters: the vision tower's cost is linear in the number of cameras, so seven cameras cost about seven times the vision-tower time of one at every refresh. HiCAP therefore treats the sensing configuration as a decision. This section states the policy, the evidence for and against it, and what it needs from data that does not yet exist. The research stream behind it (repository manifest in §9) also produced negative results that changed the design, and they are reported as such.

6.1 Camera and token policy

Frames, not cameras, are the first lever. A driving VLA that feeds four cameras with four frames each reads ≈ 2,600 visual tokens per tick (≈ 19.5 TFLOP for a 1.7 B language model, [E]). HiCAP feeds *one frame per camera* to the backbone and gives temporal information to the motion adapter at low resolution (§4.1); one to three cameras then cost 1.3–3.6 TFLOP.

Tiers. The front-wide camera is always on at 160 tokens. A tier adds cameras at 64 tokens each: T1 adds one camera (front-tele for high-speed following and distance keeping; the cross camera on the turn side for junction approach; a rear-side camera for lane-change intent and merges); T2 adds two; T3, all seven, is a probe and degraded-gate state, not a steady state. With peripheral cameras at 64 tokens, all seven fit a 100 ms tick on a 2 B-class stack (≈ 40 ms) and marginally on a 4 B-class one (≈ 68 ms), against ≈ 120 ms for seven cameras at 160 tokens ([E], scaled from NVIDIA's published rows, ±25%). The first extra camera's language-model tokens are nearly free because the language stage is weight-streaming-bound at these lengths; its vision-tower pass is not, so savings made *before* the tower (resolution, update rate, on/off) matter more than pruning after it. A scheduled *slow lane* encodes the rear and cross cameras at 2 Hz with tower and pooler only (no language-model pass) into a small peripheral-risk head, at ≈ 19% of the front lane's compute ([H], untested); it covers what a gate cannot see.

The gate and which cues it may use. A camera gate is a situation classifier in disguise, because it maps a situation to a camera set. The programme's rule that inference is vision-only for scenario classification therefore applies, and the gate is specified in three arms. **G-V** (primary, the only arm admissible under the rule) reads the front-wide pooled embedding, which the decision computes anyway, and the gate's own previous camera set (keep-alive state). **G-E** adds ego kinematics; it is an ablation until the PI rules on it. **G-T** adds the previous tick's tactical token; it is *not* information-disjoint, because the tactical decision would choose which pixels the next tick's goal path sees, so attribution between goal and gate would be lost and errors could lock in. G-T is admissible only if it beats G-V/G-E by the pre-registered margin and passes a *config-invariance* test (the goal head scored on fixed-camera states must not degrade relative to gated states) and a *gate-shuffle* test. Labels may use everything: the relevance of a camera is the nearest dynamic agent

in its field of view within a distance or time-to-collision threshold, from the 3-D agent tracks, combined with egomotion-derived manoeuvre intent.

Training. Because every camera can be cached offline, the per-option error $E_j(x)$ of the frozen backbone plus heads is computable for every training frame, and the gate minimises $\sum_j p_j(x)\,[E_j(x) + \lambda\,\mathrm{cost}_j]$ (full-information, cost-sensitive learning), with a calibrated situation head as an auxiliary or ensemble member. Errors are weighted by situation criticality: a wrong camera is nearly free in cruise and costly at a junction. The heads are trained with a camera-dropout curriculum (illustratively 40% the situation's oracle camera, 20% oracle plus one random extra, 15% all, 25% front plus a random zero to two) and are *budget-conditioned*: a configuration embedding is appended to the pooled state and the full-camera embedding is a distillation target for reduced configurations,

$$\mathcal{L}^{\mathrm{cfg}} = \sum_{c} w_c \sum_{d \in \{64,128,256\}} \mathrm{InfoNCE}\big(f_d(\mathbf{e}_c), \mathbf{a}_d\big) + \beta\big(1 - \cos(\mathbf{e}_c, \mathrm{sg}\ \mathbf{e}_{\mathrm{full}})\big). \tag{9}$$

The backbone is frozen, so this nesting lives in the trainable pooler and heads.

Dynamics: asymmetric keep-alive, shadow-mode prefetch. The gate escalates in one tick and de-escalates only after three ticks with a margin of 0.10; it never smooths symmetrically. A newly enabled camera has no temporal cache, so it is switched on in *shadow mode* (encoded, not used) for two ticks before its first use, driven by intent such as the route side.

What is not done, and why. Confidence-triggered escalation to all cameras and hedging with the runner-up camera are *not* part of the design, because they measured as useless in the toy below. Matryoshka embedding *dimensions* are adopted as a free option and are expected to save nothing: scoring 10^5 cached candidates at 256 dimensions is $\approx 2.6 \times 10^7$ multiply-adds, far below a millisecond next to a 17–120 ms encoder. The nesting that carries compute is over cameras and tokens.

6.2 Evidence: a toy world, its cliff, and what it does not show

A synthetic world with seven cameras, seven sticky situations and exactly one informative camera per situation ([M], `rd_camera_gating_toy.py`; three seeds, a cost-sensitive gate, closed-loop keep-alive and a cold-start penalty; every constant is a chosen assumption) gives orderings, not levels. Decision accuracy and cost in camera units: Six readings. (i) A gate can serve 95% of decisions at 1.64 camera units instead of 7, but

Table 3: Camera-gating toy (mean of three seeds; [M]). Only orderings are evidence; the world scores one binary decision and chooses its own constants.

policy	cost (camera units)	decision accuracy
fixed front-wide only	1	0.732
fixed all seven	7	0.936
ceiling: all seven plus true situation given to the head (reference)	7	0.976
oracle gate, cold start	1.65	0.967
learned gate (ego cues, keep-alive)	**1.64**	**0.951**
vision-only gate	1.78	0.812
vision-only gate with its own previous choice fed back	1.72	0.804

its apparent lead over fixed-all (+1.4 to +1.8 points, paired, per seed) is a head-routing-difficulty artefact: fixed-all must learn to route from noisy cues. The honest references are the oracle gate (−1.6 points) and the ceiling (−2.5 points). (ii) There is a *cliff*: accuracy is flat between 1.6 and 2.0 units, then falls to 0.915 at 1.49 and 0.825 at 1.25. (iii) A wrong camera costs almost nothing in cruise, about 22 points for the tele case and 42–46 points (chance) whenever the needed cross or rear camera is off; gate errors are *confident*, concentrated on the tele-versus-wide ambiguity. (iv) Cue admissibility decides the design: the vision-only gate reaches only 0.812, and with its own previous choice fed back it *locks in*, holding a wrong camera set for ≈ 49 ticks (≈ 5 s); it matches the ego-cue gate only if the always-on front embedding separates situations about three times better than in the base world, which is an empirical property of the frozen VLM on real frames and the first thing to measure. (v) Symmetric smoothing halves flicker but costs 0.5–0.6 points against no smoothing; asymmetric keep-alive gains +1.4 points for +6% cost; a cold start costs 0.55–0.8 points and prefetching two ticks ahead recovers 0.45. (vi) Confidence escalation buys at most +0.02 points for +28%

to +160% cost. The toy cannot say anything about real cameras: one informative camera per situation, uniform camera cost, an ad-hoc cold-start model and a chosen situation-code strength (its conclusion about vision-only gates depends on it).

Extra cameras are a data project. Every cache the programme has is front-wide only. Downloading one more upstream camera for the training set is 321 GB (front-tele) to 497 GB (cross-right) of transient volume, and all six is 2.55 TB ([E], scaled from measured chunk sizes; ≈ 9 h of build on a pod). A side-study over ∼ 30 dense chunks costs 49–388 GB but is not the parity corpus. So the *first* experiment is not the gate but the value of an oracle camera by manoeuvre, before any cache is bought: if no stratum shows an oracle camera beating front-only beyond the harm margin, elasticity is retired and front-only is shipped, which is also the cheapest outcome (H-HC10).

6.3 Audio in the shared embedding space

Sirens and horns are decisive cues that may be invisible until late, and the leading model families are becoming omni-modal. We add audio as tokens in the backbone's own space, under a stricter regime than vision because audio cannot yet be validated as driving skill.

Pattern: a gated residual, not audio inside the language model. A frozen audio tower (a BEATs-base-class encoder of ≈ 90 M parameters, or an ASR-class encoder if the chosen VLM family ships none) reads a two-second window; a two-layer Perceiver pools it to $k = 8$ tokens; and the level-query outputs are updated by one gated cross-attention read,

$$\mathbf{q}'_\ell = \mathbf{q}_\ell + \tanh(g_\ell)\,\mathrm{CrossAttn}\big(\mathbf{q}_\ell,\ \text{audio tokens}\big), \qquad g_\ell \equiv 0 \text{ at initialisation}, \tag{10}$$

so that audio-null is *bit-exactly* the original model. That is the parity property the programme needs (a modality can be added without disturbing any earlier result), and it gives a deterministic test. The audio state is refreshed at 0.5 s hops (≥ 2 Hz), enough for a siren audible at 70–100 m and an emergency vehicle closing at 40–60 m/s (1.2–2.5 s of warning, [E]). Soft audio tokens inside the language model (+25 tokens, ≈ +6% of a one-camera tick) are the ablation. The compute is negligible: ≈ 18 GFLOP per two-second window for a BEATs-class tower (≈ 0.7 ms on Thor, [E]) against 17–31 ms for the first camera.

Alignment to language and vision through the frozen text path. The gate parameters are trained so that the audio-conditioned embedding matches the frozen VLM's own embedding of the scene together with a *caption of the sound event*, $\mathcal{L}^{\text{aud}} = 1 - \cos\big(\mathbf{e}',\ \mathbf{e}_{\text{teacher}}(\text{image, “siren approaching from behind, about 40 m”})\big)$, with the caption produced from the event generator's metadata (class, bearing, distance bin, approaching), *not* from any driving label. Audio is thereby tied to language without audio–text language-model training, and the echo of a label is avoided (below). Optional pre-alignment uses public audio-caption sets. Direction of arrival needs a multi-microphone front end (inter-channel phase and level) before the tower, since mono log-mel features discard it.

The label-consistency trap, and what can be tested. A real logged trajectory never yields to a *synthetic* siren, so an audio-augmented clip has no valid driving label unless the expert was already yielding, and then audio is present only where the label is “yield”: a label echo. Therefore the audio path is trained on *event metadata* only, and its behavioural effect is tested in closed-loop simulation with an acoustic layer computed offline from the simulator's actor poses (the simulator strips audio; the layer is circular, so detection realism must first be validated on real recordings). On the current corpus we can test four things: (T1) parity, audio-null equals the original embedding bit-for-bit; (T2) event heads for class, bearing and distance on public audio-visual data and injected synthetic events; (T3) usage controls on injected events, which show that the path *can* read audio when it is the only informative channel, not driving skill; (T4) no regression, so that noisy or absent audio leaves the four families unchanged (paired). The controls are audio-null, audio-shuffle matched on speed bin and noise floor, a *bearing flip* (mirrored channels flip the predicted side), a level dose-response ($0/-10/-20/-30$ dB, monotone decay), an *onset shift* (± 1–2 s from the label-consistent time), a siren with no visible vehicle and a visible vehicle without siren, and an *audio-only label probe* (an audio-only classifier of the driving label must be at chance on real clips, the vision-only-rule leak test in a new costume). Synthetic events are rendered from world state, taking an actual agent track as the emitter and applying delay, $1/r$ gain, a Doppler shift $f' = f\,c/(c - v_r)$, air absorption, interaural time and level differences over a virtual array, and a speed-dependent ego-noise floor. Modality dropout at $p = 0.3$ keeps vision-only operation intact.

Adapter or native omni backbone. Cosmos reasoners have no audio tower, so the adapter above is the default. With the size cap lifted, a natively audio-capable backbone is an alternative that puts audio tokens in the language model's own embedding space (Qwen3-Omni-30B-A3B, Gemma-4-E4B and similar; licences and Thor rows unverified except where stated in the research note). The injection-site question is registered as a head-to-head on one frozen backbone (note R-F, E-N1): audio native in the prefill, adapter at the readout, and the adapter with the cache kept intact. Outcome N (adopt native) requires a presence-AUROC gain on real clips with no loss on injected events and at least seven of eight controls passed; outcome A (keep the adapter) if the adapter wins on injected events, native fails two controls, or the control budget cannot afford three; a tie goes to the adapter, which is cheaper and keeps the cache. The margins are fixed from the three-seed spread before launch, and no outcome is a claim that audio improves driving.

Evidence and gates. Published siren detectors with roof-mounted microphones report high presence recall with median direction error near 10° and median distance error near 9 m within 10–50 m, degrading beyond ≈ 70 m (arXiv:2109.14797, [P], one-hop). No driving corpus the programme uses carries audio (PhysicalAI-AV, nuScenes, Waymo, Argoverse 2, comma2k19; the simulator strips it); the one place not yet probed is the audio stream of the shipped video container, a ten-minute check that would overturn this section if it found one. The wake-up gate (a band-energy detector that switches the tower on) has its false-positive rate measured. The PI ruled on 2026-09-29 that audio may optionally be used at inference, which settles admissibility and leaves the design constraints that "optional" implies: the model must run bit-identically without audio (T1); every family's primary result is reported audio-off and audio-on is a paired arm, so that no headline number depends on a channel no corpus can supply; the programme's vision-only rule for the situation classifier is untouched, because HiCAP has no classifier head, and the goal-path controls (NC6) apply to the audio path as to any other input; and the audio-only label probe stays mandatory, since an audio channel that predicts the label is exactly the leak the rule exists to catch.

7 Theory

This section states what the design guarantees and under which assumptions. Every proposition is elementary; the point is to make the assumptions explicit, because several failures observed in our own programme (action echo, split-mean bias, uncalibrated imagination) came from unstated ones. A red-team pass over an earlier draft found that the InfoNCE statement did not cover the masked loss the design uses, that a conformal claim needed an unavailable conditional coverage, and that a future critic was not identified; each is corrected below.

7.1 Setting

Let $s \in \mathcal{S}$ be the multimodal state (frames, ego state, optional audio). A decision is a leaf of a rooted tree $\mathcal{T}$ of depth L: $a = (v_1, \dots, v_L)$ with $v_\ell \in C(v_{\ell-1})$, where $C(v)$ is the finite set of children of node v and v_0 is the root. In HiCAP $L = 3$ (strategic action, tactical cell, trajectory entry; §4.2); the trajectory entry is a pair (path, profile) drawn from a product, which is scored coarse-then-compose (below). Each node v has a unit-norm embedding $\mathbf{a}_v \in \mathbb{S}^{d-1}$ and each level ℓ a unit-norm state readout $\mathbf{s}_\ell(s) \in \mathbb{S}^{d-1}$. The critic is $f_\ell(s, v) = \tau \langle \mathbf{s}_\ell(s), \mathbf{a}_v \rangle$ with temperature $\tau > 0$. The expert (logged) decision is drawn from $p(a \mid s) = \prod_\ell p(v_\ell \mid v_{\ell-1}, s)$.

7.2 What contrastive training estimates, and the prior correction

Proposition 1 (Per-node InfoNCE optimum, unmasked)**.** *Fix a parent u and a state s. Draw a positive child $v^+ \sim p(\cdot \mid s, u)$ and K negatives i.i.d. from a proposal q_u on $C(u)$ with $\operatorname{supp} p \subseteq \operatorname{supp} q_u$, and remove no candidate from the normaliser. Among all critics f, the InfoNCE loss $-\mathbb{E} \log \frac{e^{f(s,v^+)}}{e^{f(s,v^+)} + \sum_{k=1}^{K} e^{f(s,v_k^-)}}$ is minimised by*

$$f^\star(s, v) = \log \frac{p(v \mid s, u)}{q_u(v)} + c(s), \tag{11}$$

for any function c of s alone.

Proof. For fixed s the loss is the cross-entropy of a $(K+1)$-way classifier that must identify which of $K+1$ candidates was drawn from $p(\cdot \mid s, u)$ rather than from q_u. Its Bayes posterior that candidate j is the positive is $\frac{p(v_j|s,u)/q_u(v_j)}{\sum_i p(v_i|s,u)/q_u(v_i)}$, and the softmax over f reproduces this posterior exactly when $f = \log(p/q) + c(s)$,

since $c(s)$ cancels in the normalisation. Cross-entropy is minimised by the Bayes posterior. This is the noise-contrastive argument of van den Oord et al. [36] and Poole et al. [30]. □

Remark (masking changes the optimum; corrected). CLM additionally removes from each row the negatives that share the positive's action. Then the positive is compared only against *different* actions, the candidates are no longer exchangeable (the Bayes posterior gains a factor the critic $f(s, v)$ cannot represent), and $f^\star$ is no longer the pointwise mutual information. We do not derive the masked optimum. A tabular numerical check on the worked example below (`RT_masked_infonce_check.py`, [M]) recovers the PMI without the mask (centred critic $(-0.237, 0.332, -0.095)$ against $(-0.238, 0.322, -0.084)$) and shows that with the mask the critic picks the most frequent action in both states, so that the $\beta = 1$ correction over-corrects (cruise probability 0.97 and 0.99 against the true 0.60 and 0.80); no single β repairs it. HiCAP's default loss is therefore the *level-wide soft-target softmax* (§4.3), a classifier whose optimum is $\log p + c(s)$ with no prior term, in which near-positive nodes receive target mass and are *not* removed from the normaliser. Masked InfoNCE, unmasked InfoNCE and the softmax are compared in H-HC4; for the masked variant β is empirical.

Corollary 2 (Selection bias and its correction, unmasked loss)**.** *If q_u is the marginal $\bar{p}_u(v) = \mathbb{E}_{s|u}\, p(v \mid s, u)$ (in-batch negatives from batches grouped by parent), then* $\arg\max_v f^\star(s, v) = \arg\max_v \log \frac{p(v|s,u)}{\bar{p}_u(v)}$*, the pointwise-mutual-information maximiser. It* can *differ from the likelihood maximiser* $\arg\max_v p(v \mid s, u)$*, and only if $\bar{p}_u$ is not uniform. Adding $\beta \log \hat{q}_u(v)$ with $\beta = 1$ recovers $\log p(v \mid s, u) + c(s)$. Because $\hat{q}$ and the critic are estimated, β is fitted on held-out training episodes rather than fixed.*

Example. Prior $(0.70, 0.20, 0.10)$ over {cruise, follow-lead, lane-change}, true conditional $(0.60, 0.30, 0.10)$ in a given state: PMI is $(-0.15, +0.41, 0)$, so the PMI argmax is "follow-lead" while the most likely action is "cruise". (Under the prior $(0.5, 0.3, 0.2)$ a conditional such as $(0.9, 0.05, 0.05)$ gives the same argmax for both: the effect needs a conditional that is not much more peaked than the prior.) Our corpus is dominated by straight cruising, so the effect is plausible; §9 gives its measured size on a synthetic world.

Proposition 3 (Tree factorisation)**.** *Let $\hat{p}(v \mid u, s) = \mathrm{softmax}_{v' \in K(u)} \big(f_\ell(s, v') + \beta_\ell \log \hat{q}_u(v')\big)_v$ over a non-empty kept set $K(u) \subseteq C(u)$ (if the rule set empties it, fall back to the top-scoring child). Then $\hat{p}(a \mid s) = \prod_{\ell=1}^{L} \hat{p}(v_\ell \mid v_{\ell-1}, s)$ is a normalised distribution over the retained leaves. By the chain rule the marginal of a node is the product along its path and equals the sum of the leaf probabilities beneath it. Two distributions coexist in HiCAP: the conditional product $\hat{p}$, used by the decision rule, and the level-wide marginals $\tilde{p}_\ell$ used to rank nodes across parents; they are separate parameters, tied by a KL term (§4.3), not equal, and the conformal keep-sets of rule R6 are calibrated on $\tilde{p}_\ell$.*

Proof. Each factor is a softmax on a non-empty set of children of one node, so it sums to one; the product over a root-to-leaf path therefore sums to one over the retained leaves by induction on depth. □

Proposition 4 (Marginal node score)**.** *Fix a level ℓ with node set V_ℓ and a beam of width w. The probability that the beam contains the ancestor of the expert's action is $\sum_{v \in beam} P_\ell(v \mid s)$, where $P_\ell(v \mid s) = \sum_{a \in \mathrm{subtree}(v)} p(a \mid s)$ is the marginal mass of the node. It is maximised over all beams of width w by the w nodes of largest* marginal*, and no node score that is not monotone in the marginal can do better at that level.*

Proof. The expert's action falls in the subtree of exactly one node of the level, so the retention probability of a set of nodes is the sum of their marginal masses; the maximum over sets of size w is attained by the w largest terms. □

This is why level critics are trained as marginals with level-wide negatives (§4.3): a sibling-only softmax leaves a per-(state, parent) offset free, and a mean-pooled child centroid scores the average child, neither of which is monotone in the marginal. Two cautions: retaining the ancestor of the single *most likely* leaf would call for a max-heap score instead (the expert's action is a draw from p, so the marginal is the right target here), and per-level optimality does not compose across levels (a parent of mass 0.5 spread over ten children beats a parent of mass 0.4 on one child at a width of one, yet retains only 0.05 of the leaf mass against 0.4); a cascade-optimal score would use the best mass reachable under the downstream widths.

Proposition 5 (Beam miss under noisy level critics)**.** *Let a level have N children, let the true ancestor have score margin m_j over competitor j, and let node scores carry independent errors of standard deviation σ. With beam width w the ancestor is lost only if at least w competitors outscore it, and*

$$P_{miss} \;\le\; \frac{1}{w} \sum_{j=1}^{N-1} \bar{\Phi}\Big(\frac{m_j}{\sigma\sqrt{2}}\Big), \tag{12}$$

with $\bar{\Phi}$ the standard normal tail. Over L levels the ancestor path survives with probability at least $1 - \sum_\ell P_{miss,\ell}$. To keep level ℓ at miss $\le \alpha_\ell$ it suffices that $w_\ell \ge (N_\ell - 1)\,\bar{\Phi}\big(m_\ell/(\sigma_\ell\sqrt{2})\big)/\alpha_\ell$ when the margins are equal.

Proof. The difference of two independent errors is normal with variance $2\sigma^2$, so competitor j outscores the ancestor with probability $\bar{\Phi}(m_j/(\sigma\sqrt{2}))$. The number of outscoring competitors has this expectation summed over j, and Markov's inequality gives $\Pr(\text{count} \ge w) \le \mathbb{E}[\text{count}]/w$. The union bound over levels gives the survival statement. □

The width is thus a proxy for critic quality: a critic that delivers margin m at noise σ buys a narrower beam, which is the rationale for a beam width that follows a calibrated margin rather than a constant.

7.3 Cost and memory of hierarchical retrieval

Proposition 6 (Retrieval cost)**.** *Let level ℓ have branching $B_\ell = \max_u |C(u)|$ and beam width b_ℓ ($b_0 = 1$). Flat scoring of $N = \prod_\ell B_\ell$ leaves costs Nd multiply-accumulates and streams $N\,d\,w$ bytes of embeddings (w bytes per element) per decision. Beam retrieval touches $M = \sum_\ell b_{\ell-1} B_\ell$ nodes, i.e. Md multiply-accumulates and Mdw bytes. For a product vocabulary of K_P paths and K_V profiles ($N = K_P K_V$ composed entries), scoring the factors separately, keeping (k_P, k_V) and composing costs $M = K_P + K_V + k_P k_V$ rows. This selects the best composed entry* exactly *only if the composed score is separable, $f(p,q) = f_P(p) + f_V(q)$; with a bounded pair interaction $|b(p,q)| \le B$, keeping every path within $2B$ of the k_P-th path score and every profile within $2B$ of the k_V-th profile score makes the composition exact (a threshold-algorithm argument). Because HiCAP's fine stage adds a pair head, the recall of composition is otherwise* measured*, not assumed (H-HC3).*

For $1{,}024 \times 256$ entries with $(20, 10)$ kept this is 1,480 rows instead of 262,144, a factor of 177 (Table 8). On an edge device the binding constraint is memory bandwidth, not arithmetic (§8), and at this size flat scoring is itself affordable (≈ 1 ms), so the case for the hierarchy is the small set the backbone can reason over and the level-wise priors, and only beyond a few million entries the speed.

7.4 Recall under cascaded pruning

Pruning is safe only if it rarely removes the right action. We give two mechanisms.

Proposition 7 (Conformal cascade)**.** *For level ℓ let $\rho_\ell(s, v)$ be any score and $\hat{t}_\ell$ the $\lceil (1-\alpha_\ell)(n+1) \rceil$-th smallest value of $-\rho_\ell(s_i, v_i^\star)$ over n calibration units, each scored against the children of* its own logged parent*. For a test window let $K_\ell^\star(s) = \{v \in C(v_{\ell-1}^\star) : -\rho_\ell(s, v) \le \hat{t}_\ell\}$ be the keep-set that would be used if the logged parent had been retained. If calibration and test units are exchangeable, then $\Pr\big(v_\ell^\star \in K_\ell^\star(s)\big) \ge 1 - \alpha_\ell$ for each level. The cascade retains the logged leaf exactly when $v_\ell^\star \in K_\ell^\star(s)$ at every level, hence*

$$\Pr\big(\textit{logged leaf retained}\big) \;\ge\; 1 - \sum_\ell \alpha_\ell. \tag{13}$$

Proof. Level-wise coverage is the split-conformal guarantee [1]: the test nonconformity $-\rho_\ell(s, v_\ell^\star)$ is exchangeable with the calibration values, so it is at most their $\lceil (1-\alpha_\ell)(n+1) \rceil$-th smallest with probability at least $1 - \alpha_\ell$. If the logged ancestors were all retained, the cascade's candidate set at level ℓ is the set of children of the logged parent, thresholded at $\hat{t}_\ell$, i.e. $K_\ell^\star(s)$; so the logged leaf is retained iff no level misses. The union bound over the L events $\{v_\ell^\star \notin K_\ell^\star(s)\}$, each of probability at most α_ℓ, gives (13). □

We state the union bound only: a product $\prod_\ell (1 - \alpha_\ell)$ would require coverage *conditional on* the parent having been retained, which conditions on an event of the test window and is not covered by exchangeability. Cascades of conformal sets with a summed miss budget are known [9]. Three cautions follow from the measurements of §9 and the red-team pass. (a) *The exchangeable unit is the episode*, not the window: windows within an episode are strongly dependent, so calibrate with one window per episode or with cluster (episode-level) conformal, on episode-disjoint splits. (b) The guarantee is marginal over calibration draws: conditional on one calibration set, coverage follows a $\text{Beta}(k, n{+}1{-}k)$ law whose median is the nominal level, so about half of all draws fall below nominal at any n even with exact exchangeability; a guarantee *per calibration draw* needs the training-conditional (PAC) form, which picks the quantile index k' so that $\text{BetaCDF}(1{-}\alpha; k', n{+}1{-}k') \le \delta$ [37]. (c) Marginal coverage hides rare classes, so keep-sets are calibrated per tactical class (Mondrian).

Proposition 8 (Physical mask)**.** *Let $\mathcal{B}$ be a set of kinematic limits (longitudinal acceleration and braking, lateral acceleration) and $\mathcal{K}_{\mathcal{B}}(s) = \{a : \text{ the implied path satisfies } \mathcal{B} \text{ given the current speed}\}$. If the logged path satisfies $\mathcal{B}$, the mask retains the* logged future *with probability one. The oracle-in-set error over the candidate set is unchanged if and only if the oracle-best* candidate *is feasible; a sufficient condition is that the limits are widened by the feasibility functional's Lipschitz constant times the quantisation radius of the vocabulary.*

That the mask retains the logged future is a statement about the bounds, not about the network, so it is checked empirically rather than proved. Section 9 reports that physical limits keep 881/881 logged 2 s futures of the validation set (one-sided 95% lower bound $0.05^{1/881} = 0.9966$ on the window-level retention rate if windows were independent) while removing 78% of a speed-agnostic anchor set, and that the oracle-in-set error is unchanged to three decimals (the direct evidence for the second claim). Tight comfort limits (3/6/4 m/s^2 for acceleration, braking, lateral) are *not* safe as hard masks (they removed 2.4% of logged futures) and are therefore used as costs. The accelerations are 0.5 s finite differences of four waypoints, which low-pass the peaks: the test is *lenient* (it under-estimates peak acceleration and errs toward retaining), not conservative for feasibility. The 40 validation episodes make the effective sample smaller than the window count, which is why the calibrated conformal mask, not this bound, is the shipped guarantee.

7.5 Predictive imagination that is identifiable

A world model trained on expert logs cannot learn what happens under an action the expert never took, because in the log the action is a function of the state. We therefore split the outcome into a part the ego cannot change over the horizon and a part it controls.

Proposition 9 (Identifiable outcome factorisation)**.** *Let $W_{t+1:t+H}$ be the future scene (other agents, signals) and $e_{t+1:t+H}$ the planned ego path. If no component of the scene is causally downstream of the ego action over the horizon (non-reactivity), then $p\big(W_{t+1:t+H} \mid s, \mathrm{do}(a)\big) = p(W_{t+1:t+H} \mid s)$, which is identified from the log by marginalising over the logged action; the observational independence $W \perp a \mid s$ is neither required nor implied (an expert acting on cues absent from s can break it without reactivity, and it can hold under reactivity by cancellation). The planned ego path $e = \kappa(a, s_{ego})$ is a known function; tracking error lies outside the proposition. Any interaction cost $c(W, e)$ can then be evaluated for every candidate a without a learned counterfactual model.*

Proof. Non-reactivity means the scene is a non-descendant of the action in the causal graph, so intervening on a leaves its conditional law given s unchanged, and $p(W \mid s)$ is a functional of the observational joint distribution obtained by marginalising the logged action. The planned ego path is a deterministic function of the candidate and the ego state. □

The assumption fails for reactive agents (a driver yielding to the ego). We do not hide this: reactivity is bounded by a learned correction term whose weight is limited and which is reported separately, and imagination is never allowed to *raise* a candidate's standing except through a learned benefit gate and a disagreement veto (§5).

Proposition 10 (Outcome critics: what is and is not identified)**.** *Let $g(s, a, x) = \tau\langle \Phi(s, a), \mathbf{z}(x)\rangle$ be trained with InfoNCE over* outcomes *x (negatives are outcomes of other windows). Its optimum is $g^\star = \log \frac{p(x|s,a)}{p(x)} + c(s, a)$, a density ratio of the dynamics, but only up to an arbitrary function $c(s, a)$ of the candidate. Hence g can rank outcomes for a fixed (s, a) but* cannot rank candidates at a fixed state*. A candidate-ranking critic must normalise over candidates for a realised outcome, a softmax over the per-state counterfactual set, whose optimum is $\log \frac{p(a|s,x)}{q(a|s)} + c(s, x)$, and this requires counterfactual data.*

Proof. The first statement is Proposition 1 with the positive drawn from $p(x \mid s, a)$ and negatives from the marginal $p(x)$: the softmax normalises over x for fixed (s, a), so any $c(s, a)$ cancels. The second statement is Proposition 1 with the roles of candidate and outcome exchanged. □

Consequently HiCAP never ranks candidates by an outcome-InfoNCE score. Outcome InfoNCE is used only as auxiliary representation shaping, with targets in the current ego frame; candidate costs come from cost heads trained on counterfactual costs where those exist (§5).

7.6 Distilling the selector into a small student

Proposition 11 (Decision preservation, single-level argmax)**.** *Let student embeddings satisfy* $\|\hat{\mathbf{s}} - \mathbf{s}\|_2 \le \delta_s$ *and* $\|\hat{\mathbf{a}} - \mathbf{a}\|_2 \le \delta_a$ *(unit vectors). Then* $|\tau\langle\hat{\mathbf{s}}, \hat{\mathbf{a}}\rangle - \tau\langle\mathbf{s}, \mathbf{a}\rangle| \le \tau(\delta_s + \delta_a)$*, and the student's arg max equals the teacher's whenever the teacher's top-1/top-2 score margin exceeds* $2\tau(\delta_s + \delta_a)$*. With shared action embeddings* ($\delta_a = 0$) *it suffices that the margin to each competitor* v *exceeds* $\tau\,\delta_s\,\|\mathbf{a}_{top} - \mathbf{a}_v\| \le 2\tau\delta_s$*.*

Proof. $\langle\hat{\mathbf{s}}, \hat{\mathbf{a}}\rangle - \langle\mathbf{s}, \mathbf{a}\rangle = \langle\hat{\mathbf{s}} - \mathbf{s}, \hat{\mathbf{a}}\rangle + \langle\mathbf{s}, \hat{\mathbf{a}} - \mathbf{a}\rangle$, each term bounded by Cauchy–Schwarz. A rank change between two candidates requires their scores to move toward each other by at least the margin, i.e. each by at least half of it. With $\delta_a = 0$ the change of the gap between two candidates is $\tau\langle\hat{\mathbf{s}} - \mathbf{s}, \mathbf{a}_{\text{top}} - \mathbf{a}_v\rangle$, bounded by $\tau\delta_s\|\mathbf{a}_{\text{top}} - \mathbf{a}_v\|$. □

The guarantee covers the single-level dot-product argmax only. For the cascade it must hold at each level's b-th/$(b{+}1)$-th boundary and at each conformal threshold, and it does not cover the re-score, the cost heads or imagination. At the CLM temperature cap $\tau = 100$ it requires $\delta_s < \text{margin}/200$, so it is close to vacuous unless margins are large; H-HC9 measures the share of real windows that meet it. Because the interface between backbone and heads is a fixed-dimension embedding space, the frozen multi-billion-parameter backbone can, if a latency tier fails, be replaced by a smaller student trained to match the cached teacher embeddings and candidate scores, and Proposition 11 converts an embedding-error budget into a decision statement for that argmax. Under the size ruling of §8.3 the student is an option, not a requirement.

8 Efficiency: what is cached, what runs when, and how large the backbone may be

Compute efficiency is a design constraint, not a tuning step. Four mechanisms carry it: multi-rate execution, caching at build time, hierarchical retrieval, and pruning before expensive computation (elastic sensing, §6, is the fifth). This section states each cost in closed form or from published figures and separates that clearly from anything we have timed. **Nothing in this section is a HiCAP timing**: rows are arithmetic ([E]) or NVIDIA-published figures for the backbone alone ([P]).

8.1 Multi-rate execution

The decision levels have different natural rates, and so does the cost. Table 4 is the schedule. The fusion

Table 4: Multi-rate schedule. The frozen VLM is the only expensive component and runs at the slowest rate; the fast path is small enough that launch overhead, not arithmetic, dominates it.

component	rate	recomputed	cost model
frozen VLM (ViT + prefill), selected cameras	2 Hz	visual tokens at three depths, last-token state; between refreshes tokens are reused with an age embedding	Table 5
motion adapter	10 Hz	16 motion tokens from the 3-frame stack	≈ 2 M params
fusion transformer	10 Hz	level queries over ≈ 160 tokens	≈ 19 M params, ≈ 8 GFLOP[E]
L1 readout / retrieval	every 2 s	strategic scores, keep-set	5 nodes
L2 readout / retrieval	every 0.5 s	tactical cell scores, classes then members; survivors cached for five ticks and re-masked each tick	≤ 35 cells; 1,280 factor rows
L3 compose, re-score, refine	every 0.1 s	compose ≈ 200 entries, late-interaction shortlist, cross-attention re-rank of $K = 16$, refinement	≈ 0.34 GMAC, Table 8
System-2 verifier	only if uncertain	one extra prefill on ≤ 5 renderings	one prefill row of Table 5

cost of ≈ 8 GFLOP is $2 \times 19\text{M} \times 200$ tokens, ignoring attention; at the ≈ 25 TFLOP/s eager-bf16 throughput class of the programme's encoder measurements[I] this is well under a millisecond of arithmetic, so the fast path is expected to be launch-bound, and the only honest statement is that it must be measured (hypothesis H-HC11).

8.2 Backbone cost on the target SoC

NVIDIA publishes measured Jetson Thor figures for Qwen3-VL-2B, -4B and -8B (TensorRT Edge-LLM 0.10.0, batch 1, vision tower FP16, language model NVFP4)[P]: for the 4 B model a ViT pass of 11.6 ms and a 292-token prefill of 22.3 ms, and a 2,048-token prefill of 62.2 ms. Interpolating linearly in token count and scaling the ViT with the number of camera images gives Table 5; the research stream, using a reduced vision-tower cost and 160 tokens per camera, estimates ≈ 31 ms and ≈ 61 ms for the Qwen3.5-4B external reference with one and three cameras (±25%, [E]); the table's row for it uses the published rows directly and is the more conservative. The ViT row is measured on a ∼265-token image while our cylindrical crop yields 160 tokens per camera, so the ViT column is an upper bound; the one-camera column ($n = 224$) lies below the smallest published prefill length (292), so it is an extrapolation. Two conclusions follow, both estimates.

Table 5: Estimated backbone latency per VLM refresh on Thor from the published rows (linear interpolation; ms). n = total tokens (160 per camera + 64 prompt/query).

	1 camera (n=224)		3 cameras (n=544)		7 cameras (n=1184)	
model	ViT	total	ViT	total	ViT	total
Qwen3-VL-2B	11.4	23.5	34.2	49.1	79.8	100.3
Qwen3-VL-4B	11.6	32.4	34.8	62.8	81.2	123.8
Qwen3.5-4B (external reference)	10.9	40.0	32.7	73.4	76.3	140.2
Qwen3-VL-8B	15.7	45.0	47.1	89.6	109.9	178.8

First, a single camera at a 2–5 Hz refresh is cheap (at 10 Hz the Qwen3.5-4B reference exceeds the duty rule) (a 4 B model at ≈ 32–40 ms), so the case for a frozen VLM in the loop does not rest on the model being slow. Second, the cost is dominated by the vision tower and is linear in the number of cameras, so seven cameras cost ≈ 124–140 ms per refresh and cannot run at 10 Hz; the value of choosing cameras per tick (§6) is exactly this slope. An open question is the precision drift between NVFP4 features on the vehicle and the BF16 features a head was trained on; it is registered as a measurement (H-HC11).

The same arithmetic explains why a decode-based VLA behaves differently: Cosmos3-Edge's 4.8 GB of BF16 weights over 273 GB/s give a 57 tok/s ceiling and NVIDIA measures 37–43 tok/s[P]; a trajectory of a few hundred tokens costs seconds, while prefill plus a table lookup costs tens of milliseconds.

8.3 Latency, not parameter count, sets the backbone size

The PI ruled on 2026-09-29 that the programme's sub-300 M cap does not bind this arm if the inference time of the whole system is very low. That turns a size rule into a latency rule, and two clocks have to be told apart because they respond to backbone size in opposite ways. *Command latency* L_{cmd} is the time from the newest camera frame to the emitted command. It runs through the motion adapter, the fusion transformer, retrieval and the heads; the backbone enters only through the cache, so to first order L_{cmd} does not depend on the backbone's size. *Token age* A is the age of the cached VLM tokens when the fast path reads them; with refresh rate f_r, refresh compute time T_r and an overhead o for pooling, hand-off and pre-processing, the worst case is $A = T_r + o + 1/f_r$. A larger backbone raises T_r, hence A and the refresh duty $D = f_r(T_r + o)$. We register three criteria in advance: $D \leq 0.5$ (the remainder is for the fast path and the vehicle's other tenants; an assumption), the fast-path p95 ≤ 25 ms *measured while a refresh is running*, and L_{cmd} p95 ≤ 100 ms, the tick. The synchronous reading, in which the refresh sits inside the tick, is the special case $f_r = 10$ Hz. The size ruling is therefore met by construction only if two things hold that no arithmetic here can show: that the refresh does not inflate the fast path through contention for the SoC's 20 streaming multiprocessors[I] and its 273 GB/s[P] (no preemption is assumed), and that stale tokens do not cost decision quality (H-HC11, H-HC15).

Table 6 gives, for the dense Qwen3-VL lineage that carries the published rows, the largest body that meets $D \leq 0.5$ under the pessimistic effective throughput. Inside the measured body range (1.4–7 B) the time is interpolated between the three published rows; above it, the eight-billion row's effective rate is assumed to be a lower bound on throughput (monotone in size across the three published points, 65/95/126 TFLOP/s at $n = 292$ and 206/239/272 at $n = 2{,}048$; not true across families, since the Qwen3.5 hybrid is about 1.5× slower per token). Entries beyond twice the largest measured body (13.9 B) are stated only as "at least", with no fitted exponent and no point estimate. The reading is direct. At 2 Hz an 8 B-class backbone (the estimate from the published Qwen3-VL-8B rows: 223.4 ms for three cameras with three frames each, duty 0.48 with the assumed 15 ms overhead) is admitted, and with one frame per camera a body of at least 13.9 B and, under the optimistic throughput only, up to about 30 B meets the duty rule at three cameras (a bracket outside the measured range, not an estimate); at 10 Hz only a 4 B-class model with one camera does, so a 10 Hz

Table 6: Largest dense backbone body (billions of parameters) meeting refresh duty $D = f_r(T_r + 15\,\text{ms}) \leq 0.5$ at the pessimistic throughput ([E], `hicap_latency_budget.py`; anchored on [P] Thor rows, ViT and LLM as in Table 5). "> 13.9": the bound lies beyond twice the largest measured body. "none": no body size in the grid meets the budget at the assumed vision-tower time. The 10 Hz one-camera entry (4.0 B) moves with the assumed tower time (about 3.2 to 4.6 B). Frames per camera is the number of images the VLM reads per refresh; the motion adapter supplies temporal information, so one frame per camera is the primary setting.

refresh f_r	1 camera, 1 frame	3 cameras, 1 frame	7 cameras, 1 frame	3 cameras, 3 frames
2 Hz (asynchronous)	> 13.9	> 13.9	12.6	7.9
5 Hz	> 13.9	5.9	none	none
10 Hz (synchronous)	4.0	none	none	none

VLM refresh forces a small backbone and the ruling buys nothing there. The price of the large backbone is age: an 8 B model at three cameras (one frame) has $A \leq 90 + 15 + 500 \approx 0.6$ s, which at 15 m/s is 9 m of travel; the fusion transformer's age embedding and the motion adapter carry the fresh part of the state, but metric quantities such as the lead gap that the imagination tiers read from the cached state are the ones a stale token cannot provide (§5); the forecast horizon is therefore counted from token capture, an assumption tested by H-HC15.

Table 7: Backbones without a published Thor row, as brackets (optimistic–pessimistic ms per refresh, ViT + prefill, no overhead; [E]). Optimistic = $\max(2P_{\text{body}}n/272\,\text{TFLOP/s},$ weight floor); pessimistic scales the 8 B row by $P_{\text{body}}/6.95$; the MoE row is bounded above by the published 137.7 ms 2,048-token row of the text model. The weight floor streams every weight once at 0.7×273 GB/s. Memory is weights only at 4.5 bit (NVFP4), 8 bit and 16 bit; cache is the int8 token cache of §8 at three depths and 128 tokens for 188,640 refreshes, at the assumed hidden width (2,048, 5,120 and 2,048 for the three rows); the 32 B row assumes the Qwen3-VL-32B architecture and a SO400M vision tower (UNVERIFIED).

backbone	1 cam	3 cam	7 cam	3 cam, 3 fr	floor	weights GB (4.5/8/16 bit)	cache GB
Cosmos3-Edge reasoner, 2.4 B	28	62	130	165	7	1.4 / 2.4 / 4.8	148
Cosmos-Reason2-32B	113–147	172–238	382–419	486–510	97	18.6 / 33 / 66	371
Qwen3-Omni-30B-A3B, 3.3 B active	106–153	137–185	200–248	231–279	90	17.2 / 30.5 / 61	148

Eight-billion-class models need no bracket: Cosmos-Reason2-8B, the Cosmos 3 Nano reasoner and the Alpamayo 1.5 language model are described as sharing the Qwen3-VL-8B architecture (a model-card statement, inherited, not re-verified), so the published 8 B row applies to all three if that equivalence holds, and its cache is 297 GB at a hidden width of 4,096 (an assumed width). The 32 B rows sit at the edge of the 250 ms budget at three cameras with one frame (172–238 ms plus the 15 ms overhead) and exceed it for any further frame or camera, so they are teachers unless the camera set is small; the MoE row is bound by weight streaming, not by its 3.3 B active parameters, which is the reason a sparse model is not cheap on a bandwidth-limited SoC. Four caveats bound everything in this subsection: the ViT cost per image is the published one (a conservative upper bound at our 640-patch crop); all rows exclude engine hand-off and any contention; NVFP4 feature drift against BF16 is unmeasured; and no TensorRT build exists in our hands for any model except through NVIDIA's published rows. The on-device benchmark (`hicap_backbone_latency_bench.py`, H-HC11) is written, unit-tested with a mock backend, and has not been run.

The scale ladder and the cost of adapting the backbone. Removing the cap changes the experimental design in two ways. First, backbone size becomes a variable: the same heads on Qwen3-VL-2B, 4B and 8B (Cosmos-Reason2 provides 2 B and 8 B members of the same architecture, inherited) give a size ladder inside one family, read by matched-step ratios and never by an exponent (H-HC16). Second, the argument for freezing weakens where geometry matters: a frozen-VLM BEV head lost 6.34 mAP against its control and unfreezing recovered 10.46[P] (one-hop, vendor-reported). Adapting the backbone (low-rank adapters on the language model and the vision tower) forfeits the cache, which is what makes CLM cheap to iterate on. The arithmetic for one epoch of 188,640 refreshes at 544 tokens for an 8 B body is $4P_{\text{body}}n \approx 1.5 \times 10^{13}$ FLOP per sample for a forward pass plus activation gradients (weight gradients only for the adapters), or 2.8×10^{18} FLOP per epoch, which is 8–13 GPU-hours at an assumed 60–100 TFLOP/s effective bf16 on an A40-class part ([E]; activation memory not costed; the full-backward figure is $\approx 1.5\times$ that). This is affordable once and not for a grid, so the adapted arm is pre-registered after the frozen ladder (H-HC17), never before H-HC0.

8.4 Retrieval cost of a very large vocabulary

The trajectory vocabulary is a product of K_P paths and K_V speed profiles, so scoring the two factors separately, keeping (k_P, k_V) and composing costs $K_P + K_V + k_P k_V$ rows instead of $K_P K_V$ (Proposition 6). Table 8 compares this with scoring every composed entry, at $d = 512$ against the 273 GB/s of Thor[P]; the last column is the time to stream the flat table once at the published peak bandwidth. The honest reading is

Table 8: Flat versus factored retrieval per decision ([E], from Proposition 6); bf16, $d = 512$. Flat time is the table read at the published peak bandwidth of 273 GB/s, a single reader; the backbone competes for the same bandwidth.

$K_P \times K_V$	composed N	kept (k_P, k_V)	rows touched	ratio	flat MB (time)	factored MB
$1{,}024 \times 256$	2.6×10^5	$(20, 10)$	1,480	177×	268 (0.98 ms)	1.5
$2{,}048 \times 512$	1.05×10^6	$(32, 16)$	3,072	341×	1,074 (3.9 ms)	3.1
$4{,}096 \times 1{,}024$	4.2×10^6	$(40, 20)$	5,920	709×	4,295 (15.7 ms)	6.1

a scale argument. At the vocabulary sizes of published scorers (4,096–8,192 entries) and even at 262,144, flat scoring streams a few hundred megabytes in about a millisecond, so the hierarchy buys no speed at inference; the bandwidth argument begins beyond a few million entries, where an approximate index would otherwise be needed (none is a published 10 Hz edge precedent). The hierarchy earns its place elsewhere: it gives the backbone a *small* set to reason over (the cascade of §4.4 costs ≈ 0.34 GMAC per tick at $K = 16$, and the cross-attention re-ranker at $K = 8/16/32$ is 0.20/0.29/0.48 GMAC, [E]), it gives level-wise priors and a native tactical readout, and it estimates priors for entries with almost no data (about 1.5 training windows per leaf at this size). A failure of the tree is therefore not fatal: flat scoring is the reference arm and the fallback. Training is affected earlier than inference: a flat softmax over 262,144 entries at batch 2,048 materialises ≈ 0.5 G logits per step, whereas the level softmaxes are over a few thousand nodes. Whether a larger vocabulary *helps* accuracy is a separate, empirical question (H-HC5): an oracle-in-set curve is necessary but not sufficient, since the programme's own measurements show the selection gap growing with the candidate set for a fixed scorer (REF-C-XL restricted to 64/128/256 anchors: oracle-in-fan 0.4368/0.2624/0.1640 m while the share of windows in which the pick is at least twice as bad as the best rises 0.1771/0.3190/0.4540; registry §4)[I].

8.5 Caching at build time

Training the heads on cached features is what makes CLM cheap to iterate on. For HiCAP the dominant cache is the VLM tokens: at a 2 Hz refresh, 26.2 h of video is 188,640 VLM ticks; 128 tokens of width 2,048 (the hidden size of the Edge reasoner, read from its published configuration, and of the 2 B tier; the 8 B tier is 4,096 and doubles the figure) at three depths and one byte per element is 148 GB, and 64 tokens is 74 GB ([E], `hicap_cost_model.py`). A 10 Hz refresh would be 742 GB at int8 and is therefore not cached. Action embeddings are cached per step as a table and are tiny by comparison. Ablations over heads, losses and vocabularies then cost GPU-minutes on cached tensors rather than the ≈ 45 GPU-hours per arm that the programme's reference planner needs by arithmetic on an inherited receipt (40,284 steps at 4.0 s/step on an A40; [E], [I]); the cache is built once per backbone and once per camera set.

8.6 Backbone choice

The PI ruled that a VLM is the backbone and named NVIDIA's Cosmos repository as its source. Read as a URL, `github.com/NVIDIA/Cosmos` resolves (GitHub is case-insensitive) to the repository `NVIDIA/cosmos`, the home of Cosmos 3: same repository identifier and same head commit under both spellings[M] (research note R-E, §0.1). The earlier Cosmos-Reason repositories sit in another organisation, are marked as no longer under active development and point to it; Alpamayo is in a third. No model called "Cosmos Drive Nano", the name in the original request, exists in the model tables of `NVIDIA/cosmos`, in the twelve repositories of the other organisation or in the five Alpamayo repositories (absence at these locations, not proof). Taken strictly, the ruling selects the Cosmos 3 reasoner towers: Edge (a 4 B model whose reasoner is 2.43 B), Nano (16 B, reasoner 8.76 B) and Super (64 B, the teacher); Cosmos-Reason2 is their predecessor and, unlike them, has published Thor anchors. Whether "the Cosmos arm" means the Cosmos 3 towers or also Cosmos-Reason2 is a decision we ask the PI to confirm (§12); the ladder below includes both and says which is which.

The 4 B checkpoint and a driving version. The PI asks for the 4 B model. The 4 B Cosmos3-Edge checkpoint is a mixture-of-transformers: the reasoner tower (2.43 B) that HiCAP taps and a generator tower

(≈ 1.6 B, an estimate) that prefill-only use leaves idle; we load the 4 B checkpoint and run only the reasoner, so no latency below counts the generator. We searched for a driving-post-trained Edge and for a Nano "auto" variant: NVIDIA's pages and the technical report describe driving *capability* (road-scene understanding, intent prediction, a driving-reasoning benchmark column, AV chain-of-thought datasets, a 9-dimensional AV action format), and the published post-trained Edge examples are robot policies; we found no checkpoint named or documented as driving-tuned for Edge or Nano, but the model hub itself was unreachable, so its card list is unchecked. The PI names Cosmos3-Nano-Transfer-Auto as a driving version. Its model card (read with the PI's token) shows that it is a post-trained *generator*: a 16 B mixture-of-transformers derived from Cosmos3-Nano that synthesises synchronised multi-view camera video (up to 11 views) and LiDAR from a text prompt, world-scenario-map controls and optional LiDAR controls, under OpenMDW-1.1; its causal distillation Cosmos3-Nano-Sim-Auto is the real-time closed-loop simulator. Neither is documented as a reasoner, and whether the reasoner tower inside them was changed is unstated; they do not provide the driving-tuned *backbone* the H-HC2 test needs, but they are the natural tools for two other jobs: appearance and weather augmentation of the training scenes, and the closed-loop evaluation of H-HC14 (Sim-Auto). The Edge model card itself (read with the PI's token) settles the question for the primary: Cosmos3-Edge is a general physical-AI model that already reasons about driving scenes. Its card reports a "Driving" reasoning column beside General, Robotics and Smart Infrastructure (the scores are in an image we could not read), lists a 9-dimensional autonomous-vehicle action embodiment, and its named training sources include a private dashcam set (Nexar, 0.6 M samples) among 19 M reasoning images and 1 M reasoning videos under OpenMDW-1.1; it does not quantify the autonomous-driving reasoning data, and the technical report's driving datasets are unread. So Edge is driving-aware without being driving-tuned, and the contamination caveat applies to it too. The decision is therefore: Edge 4 B is the primary; Cosmos 3 Nano (reasoner 8.76 B) is the alternative if Edge fails H-HC0 or H-HC16; the driving-tuned test remains the Alpamayo pair of H-HC2.

Why large backbones keep the system fast. One exception: the optional System-2 verifier of §4.4 puts one backbone pass in the command path when it fires, so it is outside the claim that L_{cmd} is independent of backbone size; it is budgeted separately as a deadline-bounded call (about 58.6 ms for an 8 B body, [E]), counted in the duty of the ticks where it runs, and gated on a measured miss rate. The L_{cmd} criterion is a p95; a p99 and the share of ticks missing the 100 ms deadline are reported beside it, and a deadline miss triggers a hold of the last command.

The contrastive design is what keeps command latency low: choosing among a quarter-million trajectories is a table lookup of about 1.5 thousand rows (≈ 1 ms even if flat), not a generation, and the backbone runs once per refresh, prefill only. The numbers above are not the cost of the decision; they are the cost of one backbone refresh, which runs asynchronously and whose only effect on the driver is the age of its cached tokens. They matter for two things only: whether a refresh fits its period, and how old the tokens are when read.

What the models are. The Edge reasoner is a Nemotron-dense 2 B language model with a SigLIP2-SO400M vision tower and a 2 × 2 patch merger: 0.49 B for tower and merger, 1.41 B for the language body and 0.54 B for the untied embeddings, 2.43 B in all by arithmetic on the framework's published configuration[M]; its patch size is 16, so a 256 × 640 frame is 160 tokens, it has no audio tower and it emits text only. The Nano and Super reasoners are Qwen3-VL-8B and -32B in architecture (the framework's configuration names them), and Cosmos-Reason2 (2 B, 8 B, 32 B) and the Alpamayo 1.5 language model are stated to follow the Qwen3-VL architecture; the checkpoint configurations of those last two could not be read (the model hub was unreachable), so their identity with Qwen3-VL, which is what lets NVIDIA's published Qwen3-VL rows stand in for them, is inherited, not verified[I]. The licence of the Cosmos 3 code and weights is OpenMDW-1.1 (repository file); Cosmos-Reason2 weights carry NVIDIA's Open Model License, whose terms we could not read.

Table 9 settles what the latency rules exclude. The strict synchronous reading (refresh ≤ 50 ms) is met at one camera and one frame only by the ≤ 2.4 B tier in NVFP4; with three cameras nothing passes it, because the vision tower alone costs 21–29 ms. The asynchronous reading (refresh ≤ 250 ms) is met at three cameras and three frames by the ≤ 2.4 B tier and by the 8 B tier in NVFP4, and only the small tier reaches seven cameras. Precision is the largest lever, larger than the choice among the shortlisted models, and it is the unmeasured one for the primary (no optimised row, and NVFP4 for the Edge reasoner is not documented by NVIDIA). Audio is not part of any Cosmos reasoner: the Edge and Nano reasoners take text, images and video and the Nano reasoner "does not currently support the audio modality by design", so the optional audio channel enters through the adapter of §6.3 or through a native omni backbone.

Table 9: Backbone ladder under the latency criterion, in test order. Latency is ViT + prefill in ms for 1 camera/1 frame, 3 cameras/1 frame, 3 cameras/3 frames ([E], estimator of `RE_cosmos_backbones.md`: published Qwen3-VL Thor rows scaled by FLOPs, ViT scaled by patches; error about ±35% and biased high in its out-of-sample check on Qwen2.5-VL-7B). The estimator of Table 6 does not scale the ViT and is the more conservative: for the 8 B tier at 3 cameras and 3 frames it gives 223 ms against 167 ms here, so the two bracket the expectation from above.

rank	backbone	role	latency ([E])	why / risk
1	Cosmos3-Edge reasoner (2.43 B, OpenMDW-1.1)	primary frozen backbone	FP16 (the precision NVIDIA's CI tests) 41 / 76 / 180; NVFP4 (untested) 22 / 43 / 109	the only model NVIDIA built and benchmarked for Thor; TensorRT Edge-LLM has a documented reasoner path. *No optimised latency row exists*: NVIDIA's only Thor number is eager BF16, 0.19 s for a 911-token image prompt[P]. Not Qwen, so no same-architecture driving-tuned twin
2	Cosmos-Reason2-2B (≡ Qwen3-VL-2B)	measured-latency twin, ladder rung	NVFP4 19 / 36 / 85, from published rows by identity	the only 2 B-class model with real Thor TensorRT anchors; superseded, custom licence; identity unverified; its sample script upscales a 256×640 frame to about 286 tokens unless `min_pixels` is lowered
3	Alpamayo-1.5 language model (≈8 B) versus its parent Cosmos-Reason2-8B; Cosmos 3 Nano reasoner as a general twin	paired driving-aware arm	NVFP4 39 / 71 / 167; Alpamayo's only TensorRT path is FP16, 135 / 232 / 523	same architecture, differ by driving post-training: the cleanest test of driving knowledge, but 8 B against 8 B, not size-matched to rank 1, and presumed contaminated on our validation clips (below). Nano is not in the TensorRT Edge-LLM list (NIM only, a black box with no hidden-state tap)
4	Qwen3.5-4B, Qwen-Drive-1.0-4B (Apache-2.0)	external references	rows of Table 5	outside the PI's source ruling; the only 4 B driving-tuned pair with a same-architecture general twin
—	Cosmos-Reason2-32B, Cosmos 3 Super, Alpamayo-2-Super	teachers, labellers	NVFP4 ≥ 142 even at 1 camera (extrapolated)	never a refresh model; NIM itself calls Super not recommended on Thor

Taps and runtime. The frozen-tap design reads three depths. For the Qwen3-VL architecture the hidden-state tuple is per layer; for the Edge reasoner it is per *block*, and its 56 alternating attention and feed-forward blocks give 57 entries, so "middle, three quarters, final" are entries 28, 42 and 56, not 14, 21 and 28; the cache manifest stores the indices and their unit so that features of different backbones are never mixed. TensorRT Edge-LLM already exports multi-layer hidden states for other purposes (speculative-decoding bases and a thinker-to-talker hand-off), so a three-depth engine is a small export change (a hypothesis, untested), but truncating the language model cannot be combined with that export as shipped. The OpenAI-style servers (vLLM, NIM) expose no hidden state, so the design needs an in-process engine. A prefill-only build can drop the language-model head, saving 0.6–1.6 GB. The cache widths of §8 follow the hidden size: 148 GB at 2,048 (Edge, the 2 B tier), 297 GB at 4,096 (8 B tier), 371 GB at 5,120.

Contamination. A driving-tuned backbone may have seen our validation clips, and the evidence needs a precise reading. Matching clip identifiers against the 5,000-clip sample of the PhysicalAI-AV train split that NVIDIA publishes as the calibration set for quantising Alpamayo 1.5, 13 of 400 clips of an older clean validation split (3.25%, Wilson 95% interval $[1.9, 5.5]$) and 76 of 2,400 parity training clips (3.17%) are in that sample[M] (a lower bound; the canonical 40-episode validation set keeps only a four-character clip-id prefix and could not be tested). This proves membership of the public train split, not that Alpamayo trained on the clips; but Alpamayo reports training on 80–110 k hours against the 1.7 k hours of the public dataset[P], so overlap with any clip of ours can be neither excluded nor bounded, and the Cosmos 3 training data is undisclosed, so the Cosmos arms are not a clean control either. HiCAP's evaluation surface is the 139 evaluation clips of the v9 release, whose identifiers are stored as sha12 = $\text{sha256(clip id)}_{[:12]}$, so NVIDIA's lists can be hashed and intersected. Measured[M]: 4 of 139 evaluation clips (2.9%, Wilson 95% $[1.1, 7.2]$) and 81 of 4,369 train clips (1.85%) are in the 5,000-clip train-split sample; 0 of 139 and 3 of 4,369 are in the 644-clip public gold evaluation list. The four evaluation clips are excluded from every H-HC2 reading. The least driving-specific candidate is Cosmos-Reason2 (physical common sense, no autonomous-driving claim in its README), which is why it is the parent in the H-HC2 pair.

Two further facts from the research streams shape the protocol. (i) *No published result exists for a never-driving-tuned frozen VLM plus small heads making driving decisions.* Alpamayo's "frozen backbone" is frozen

after 80–115 k hours of driving fine-tuning, and its planner reads the key–value caches of *all* layers of the language model *after* the model has decoded its chain-of-causation text, not a prompt-time state; HiCAP's prompt-time taps on that model therefore measure the driving knowledge that survives in prompt-time states, not the deployed path. HiCAP on a raw VLM is a bet, not a replication, which is why the pre-registration starts with a gate (H-HC0) rather than with the full system. (ii) Freezing costs little for semantic tasks and a lot for geometric ones (a frozen-VLM BEV head lost 6.34 mAP against a control, and unfreezing recovered 10.46; classical visual odometry beat the best VLMs on ego-motion)[P] (one-hop, vendor-reported). This is the reason for the motion adapter and for feeding ego kinematics as numeric tokens at the fusion stage, and for exposing three depths of the backbone rather than the final layer.

8.7 The student, now an option

Before the size ruling of §8.3 every backbone in Table 9 was at least 6.7 times the programme's sub-300 M target and a student was mandatory. It is now a fallback with three triggers: no backbone tier meets the latency criteria at the camera count the elastic controller needs; the fast path is inflated by refresh contention beyond its p95 budget (H-HC11); or a deployment target smaller than Thor is chosen. We distinguish *HiCAP-R*, the frozen reference, from *HiCAP-S*, a smaller student that occupies the same interface (its size is whatever the failed tier requires, no longer a fixed 300 M). Because the action tower depends only on vocabulary strings and residual profiles and never on images, the student *reuses the teacher's cached action embeddings verbatim*; only the state side is distilled. The student is trained on the cached teacher readouts $\mathbf{s}_\ell$ (cosine and mean-squared error), the per-level distributions (KL) and the cost heads, and may use unlabeled driving video, since the teacher supplies the targets. Distilling per-family sub-scores listwise rather than one scalar follows the Hydra-MDP precedent (five per-family heads 83.0 against a scalar 80.2 score in that paper's metric)[P] (one-hop). Proposition 11 turns an embedding-error budget into a decision guarantee: the student's choice equals the teacher's whenever the teacher's top-two score margin exceeds $2\tau(\delta_s + \delta_a)$, and with shared action embeddings $\delta_a = 0$. Whether that margin holds on real windows is measured (H-HC9); if it does not, the honest fallback is a student trained end to end on the same tree, which loses the guarantee but keeps the interface.

9 Preliminary measurements

No HiCAP model exists, so these are the only measured statements in the paper. They are deliberately narrow: numpy-only scripts on committed artifacts, each using only quantities available at inference, plus toy worlds in which the true answer is known. Each of them tests one *mechanism* the design relies on; none tests the system. Scripts and result files are listed in the repository manifest at the end of this section.

9.1 The unmasked contrastive optimum is biased toward rare actions, and a held-out log-prior repairs it

Corollary 2 predicts that an InfoNCE-trained selector without a false-negative mask picks the pointwise-mutual-information maximiser, which under-chooses the dominant action. We test this in a synthetic driving-like world with 60 states, 16 actions and a cruise-dominated prior (true cruise share 0.503 on average), a dual encoder of width 24 with L2-normalised embeddings and 1/0.07 logit scale as in CLM, trained for 4,000 steps at batch 256; five seeds, each a new world and a new training run. Both the unmasked bidirectional InfoNCE and CLM's masked variant (same-action negatives removed) are trained. The prior $\hat{q}$ is estimated from training draws, and the log-prior weight β is chosen on a *held-out* sample of 20,000 expert actions by empirical top-1 accuracy over a grid that brackets zero (-1 to 2); the metric is the *hit rate*, the probability that the selected action equals a fresh draw from the true $p(a \mid s)$ (the maximum-likelihood rule is the upper bound for a deterministic pick). The exact PMI critic scored by argmax is far worse than always predicting the prior (0.361 against 0.503), and the trained unmasked critic reaches that optimum (0.355; it never picks cruise): the bias is not a small-sample artefact but the optimum itself. Adding the log-prior with the held-out weight, which comes out at exactly 1 in every seed as Corollary 2 predicts, brings it to 0.585, within 0.012 of the maximum-likelihood rule and level with the soft-target classifier (0.589). CLM's mask changes the picture: the masked critic is nowhere near the PMI optimum (it picks cruise in 45% of states and scores 0.571), its best held-out β is 0.5–0.75, and it reaches 0.593. A tabular check of the masked loss on the worked example of §7 behaves differently again (it picks the most frequent action), so the masked optimum is settled by neither toy; the unmasked loss and the soft-target softmax are the reference arms of H-HC4, and masked InfoNCE is an ablation whose β is empirical. An earlier version of this toy chose β by maximising

Table 10: Toy world ([M]; `hicap_pmi_toy_v2.py`). Hit rate and share of states in which the selector picks "cruise", mean ± SD over five seeds. The true cruise share is 0.503.

selector	hit rate	cruise selected
maximum-likelihood oracle (upper bound)	0.597 ± 0.006	0.613
PMI oracle (the optimum of the unmasked loss)	0.361 ± 0.019	0.000
prior only (always the most frequent action)	0.503 ± 0.027	1.000
soft-target classifier (vocabulary softmax)	0.589 ± 0.006	—
unmasked InfoNCE, argmax	0.355 ± 0.020	0.000
unmasked InfoNCE $+\beta \log \hat{q}$, β fitted (= 1 in 5/5 seeds)	0.585 ± 0.006	0.597
masked InfoNCE (CLM), argmax	0.571 ± 0.008	0.450
masked InfoNCE $+\log \hat{q}$ ($\beta = 1$)	0.582 ± 0.003	—
masked InfoNCE $+\beta \log \hat{q}$, β fitted (0.5 in 4/5 seeds, 0.75 in one)	0.593 ± 0.004	0.587

the evaluation metric under the true probabilities and only trained the masked loss; both defects were found in review and are corrected here.

9.2 Coverage of the trajectory space by a nested vocabulary

Using the 256 farthest-point anchors of the programme's reference planner (selected from 200,000 parity-training windows) and the 881 windows of the 40 validation episodes, Table 11 gives the ADE@2 s of the *best* member of nested prefixes, with episode-cluster bootstrap intervals ($B = 2000$). The second column rescales each anchor uniformly so that its implied initial speed equals the window's current speed, a crude stand-in for the residual-over-prior parameterisation that uses only the inference-time speed. Two things follow. Coverage

Table 11: Oracle-in-set ADE@2 s (m), full-set mean with 95% episode-cluster interval; 881 windows, 40 episodes ([M]; `reff_v0_coverage.py`). This is an upper bound on what a selector over the set can achieve, not a driving result.

set size	absolute anchors	speed-normalised anchors
16	2.307 [2.111, 2.508]	0.468 [0.389, 0.554]
32	1.661 [1.533, 1.776]	0.375 [0.314, 0.444]
64	1.148 [1.090, 1.211]	0.270 [0.225, 0.318]
128	0.803 [0.760, 0.842]	0.221 [0.188, 0.259]
256	0.599 [0.564, 0.634]	0.179 [0.151, 0.207]

improves steadily with set size and has not saturated at 256 for the speed-agnostic set, which is the argument for a much larger vocabulary. And removing the speed-dependent part of the path shrinks the oracle-in-set error by 3.4× at 256 entries (a set of 32 speed-normalised anchors already beats 256 absolute ones), which is the argument for the residual parameterisation. Both are properties of the candidate *set*; the programme's own measurements show that a fixed selector does not automatically convert a larger set into a better pick (§8).

9.3 A kinematic-feasibility mask on real windows

Rule R1 of §4.4 keeps a candidate only if its implied acceleration, braking and lateral acceleration, given the window's current speed, lie inside limits. Table 12 applies three limit settings to the same 256 anchors and 881 windows, and reports the share of candidates removed, whether the *logged* 2 s future itself passes the mask (recall), and the oracle-in-set error before and after. Only four waypoints exist per path, so accelerations are 0.5 s finite differences, coarse but conservative. On the logged futures the largest observed longitudinal acceleration is 2.81 m/s^2, the largest braking 3.51 m/s^2 and the largest lateral acceleration 5.29 m/s^2 (`hicap_prior_mask_result.json`). Three findings. (i) The physical limits remove 78% of a speed-agnostic anchor set and lose nothing: every logged future passes and the oracle error is unchanged (episode-cluster intervals identical to three decimals). If windows were independent this would bound the retention rate below by 0.9966 at 95%; with 40 episodes the effective sample is smaller, which is why the calibrated conformal mask, not this bound, is the shipped guarantee. (ii) Tight comfort limits are unsafe as hard masks: they remove 2.4% of logged futures and raise the oracle error by 0.03 m, so comfort belongs in the cost, not the mask. (iii) The mask is much less powerful on a candidate set that already knows the current speed: on speed-normalised anchors it removes only 15–30%. This is the honest consequence of building feasibility into

Table 12: Mask results ([M]; `hicap_prior_mask_measure.py`). Limits are (acceleration, braking, lateral) in m/s^2. *Removed* is the mean share of the 256 anchors removed per window; *recall* is the share of windows whose logged future passes; oracle ADE@2 s (m) with the mask versus without.

limits	anchors	removed	recall	oracle, masked	oracle, unmasked
physical (6, 10, 9)	absolute	0.784	1.000 (881/881)	0.599	0.599
comfort-plus (4, 8, 6)	absolute	0.846	1.000	0.602	0.599
tight (3, 6, 4)	absolute	0.908	0.976	0.629	0.599
physical (6, 10, 9)	speed-normalised	0.146	1.000	0.179	0.179
comfort-plus (4, 8, 6)	speed-normalised	0.219	1.000	0.181	0.179
tight (3, 6, 4)	speed-normalised	0.300	0.976	0.197	0.179

the parameterisation, and it means the rule's value is largest for speed-agnostic or very large sets and smallest for the residual codebook, which is what we would predict and will report.

9.4 Conformal keep-sets on real windows: coverage holds on average, and varies a lot between calibrations

Proposition 7 guarantees marginal coverage under exchangeability. To see what that means with few episodes we ran it on the 881 validation windows, splitting the 40 episodes at random 500 times into fit (14), calibration (12) and test (14) episodes, so that no test episode is ever seen by the fit or the calibration. The decision unit is a *kinematic proxy* of the tactical cell (5 lateral × 4 longitudinal classes derived from the window's own future path with thresholds chosen for this check, not the label emitter's; 16 of 20 cells occur; the most frequent, keep-lane-and-cruise, holds 48% of windows), and the score is a train-fitted cell frequency, either unconditional or conditioned on a speed bin of the current speed. Three readings. (i) The marginal guarantee

Table 13: Split-conformal keep-set for the tactical cell, 500 random episode-disjoint splits ([M]; `hicap_conformal_cells_measure.py`). Coverage is the share of test windows whose true cell is kept. The last column is the standard deviation that exact exchangeability would give for ≈ 264 calibration and ≈ 310 test windows (Beta law of split-conformal coverage plus binomial test noise, [E]).

score	nominal	mean	SD, splits	5th pct	cells kept (of 20)	pruned	SD if exchangeable
prior only	0.95	0.957	0.046	0.870	13.2	34%	0.018
prior only	0.90	0.915	0.062	0.792	10.7	47%	0.024
prior only	0.80	0.822	0.087	0.663	7.0	65%	0.033
speed-conditional	0.95	0.957	0.051	0.857	15.3	24%	0.018
speed-conditional	0.90	0.912	0.070	0.777	11.5	43%	0.024
speed-conditional	0.80	0.815	0.093	0.646	6.6	67%	0.033

holds on average: mean coverage is at or slightly above nominal at all three levels. (ii) It is a statement about averages over calibration draws, and the spread of a single draw is 2.5–2.8 times what exchangeable windows would give: the standard deviation is 0.05–0.09 against 0.02–0.03, because windows within an episode are strongly dependent and effectively few. (The share of draws below nominal is not informative: under exact exchangeability it is close to one half for every n, and a red-team simulation gave 0.43–0.50; the spread and the lower tail are what matter.) The 5th-percentile split covers 87% at nominal 95%. A shipped mask therefore needs a calibration set of hundreds of episodes and a stated lower-tail coverage; the training split has 4,572 clips. (iii) At 95% coverage the tactical cell alone is pruned by about a third, and conditioning on speed did *not* help at this sample size (the conditional table of 6 bins × 20 cells is fitted on about 350 windows and prunes less, 24% against 34%), so R2 is not expected to earn its place until it is fitted on the full training split. This is a small val-only check with a proxy label; it tests the mechanism, not the shipped guarantee.

9.5 Hierarchical retrieval over a product vocabulary (toy world)

A synthetic world with a product action vocabulary (8 strategic × 32 cell classes × 256 members = 65,536 leaves, and a replicate with 131,072 shaped like the v7 tokens) tests the retrieval mechanics with exact critics plus independent noise ([M], `rb_hier_retrieval_toy.py`; 400 test states, noise $\sigma = 0.5$). Because the critics

are exact by construction, "tree ≈ flat" is partly built in; what the toy can show is cost, the effect of critic type, and the prior. Beyond the table:

Table 14: Beam retrieval versus flat scoring in the toy ([M]; 65,536 leaves; multiply-adds counted, not timed). Agreement is the share of states in which the pick equals the flat pick; recall is the share of states in which the expert's cell is kept.

retrieval	MACs per state	reduction	agreement with flat	cell recall
flat, all leaves	3.36×10^7	1×	1.00	—
tree, beam $(1, 4)$	5.4×10^5	61.6×	0.97	0.958
tree, beam $(2, 8)$	1.09×10^6	30.9×	0.99	0.990
adaptive calibrated-margin beam, $\alpha = 0.02$	3.4×10^5	100×	—	0.966
adaptive calibrated-margin beam, $\alpha = 0.05$	2.5×10^5	133×	—	0.927
factored 20×10 composed set	—	78×	0.95	—

(i) With marginal-consistent critics the beam is not the bottleneck, the critic is; a mean-pooled centroid index loses 5 points of cell recall at beam $(3, 16)$ where a marginal critic loses 0.45, and sibling-only negatives depress tactical agreement from 0.91 to 0.62 (Proposition 4, §4.3).

(ii) The log-prior correction is mandatory at every size tested: without it the hit rate is 0.47 of the exact maximum-likelihood rate and the cruise cell is picked in 46% of states against 66% for that rule; with it, 0.83 and 65%. The bias did not grow between 4,096 and 131,072 leaves. A prior estimated from at most 0.023 expert samples per leaf recovers only 50–75% of the true-prior gain, which is why the prior is estimated hierarchically. For a bounded critic $\beta = 0.75$ and $\beta = 1$ tie on hit rate but differ 2.9× in mean regret, so β is chosen by regret and tactical agreement, not hit rate.

(iii) A hard comfort mask (nuPlan limits) removes 47.7% of the leaves with 98.3% oracle survival on average, but survival is 99.4% for cruise and 93.3% for stop and 94.3% for turn states; inflating the limits by 0.18 lifts survival to 99.4% at the price of keeping 58.3% of the vocabulary instead of 52.3%. Marginal coverage hides the rare classes, so calibration is class-conditional and episode-blocked; the analytic box test never removed a node containing a feasible leaf (0 violations).

(iv) A re-ranker after the tree gains almost everything by $K = 8$ (hit rate 0.827 → 0.940), then 0.949 at $K = 16$ and 0.953 at $K = 32$, while the oracle error of the candidate set keeps falling. This is a toy world in two dimensions with iid states and no label noise; it does not measure oracle ADE against vocabulary size on real data (that is a CPU experiment on the training split, H-HC5), latency, agents or episode clustering.

9.6 Imagination (toy world)

A one-dimensional driving world with a hidden lead-vehicle braking mode and onset, a noisy cue through a fixed random encoder and an exact simulator scores selectors by regret against the best candidate ([M], `rc_latent_imagination_toy.py`; three seeds, paired bootstrap over pooled ticks; signal-to-noise ratio 1 and 1,200 training ticks unless stated). Some arms, including the risk-sensitive floor and the model-free value, were added after the first sweep and are disclosed as post hoc. Findings.

Table 15: Imagination toy: regret (lower is better; oracle = 0), mean over three seeds ([M]). The last rows are references, not deployable arms.

selector	regret
one-hot expert label, contrastive selector (baseline)	7.00
soft outcome-aware target, no imagination (Tier 0)	3.43
re-rank of the top four by imagined rollouts	3.79
imagined rollouts of all candidates	2.17
factorised: exogenous E-head plus analytic ego (Tier 1)	**1.81**
soft targets plus re-rank (Tier 0 + imagination)	2.28
model-free value estimate, same rows	2.97
risk-sensitive lookup that never sees the hidden consequence (floor)	2.17
factorised with the true visible state (reference)	1.37

(i) Replacing the one-hot label by a soft outcome-aware target removes more than half the regret at zero inference cost (paired 3.57, $[2.99, 4.19]$), and a lookup that never sees the consequence but is risk-sensitive

already reaches 2.17: most of the apparent value of imagination is target shape and risk sensitivity. Any imagination gain must be reported against both.

(ii) Only the factorised arm is robust to the absence of counterfactual data (1.80); unstructured imagination collapses (71.8 with 40% unsafe picks for pure rollouts, 47.0 for the model-free value, 48.2 for a distilled arm).

(iii) Imagination alone chose an infeasible over-braking candidate in 89.8% of ticks, and an analytic feasibility mask after the learned term brought that to 0%.

(iv) Heuristic triggers (need-times-trust, margin) were no better than random (35–39% of the gain at 30% of the budget); a learned gate reached 65%, and imagination hurt on 13.4% of ticks even when it helped on average.

(v) Adding an imagined-cost term flipped the correlation of confidence with regret from -0.11 to $+0.19$ in all three seeds (anti-calibration, as measured earlier in the programme).

(vi) The disagreement of two heads detected out-of-vocabulary candidates with AUROC 0.877 (recall 64% at 5% false positives) but not as a benefit trigger.

(vii) The echo control worked as intended: the ego-speed R^2 was 0.620 with the real cue and 0.613 shuffled (the ego part is predictable from the ego), while lead speed fell from 0.387 to -0.329 when the cue was shuffled.

(viii) When the cue is clean (signal-to-noise ratio 3) the factorised arm lost to the risk-sensitive floor (3.00 against 2.17) because of visible-state readout error; its gap to the true-state reference (3.00 against 0.46) is the readout error that E-I0 is designed to measure on real embeddings. Of the five pre-registered readings of the stream, one was supported but confounded (imagination beats the baseline), one was supported (an infeasibility mask is needed) and three were refuted as worded (imagination beats the floor at zero signal, gates by need or margin, and compounding of cost-level disagreement). The exact simulator supplies counterfactuals that a real system does not, there are three seeds, and nothing transfers to VLM scale.

9.7 Repository manifest for this section

artifact	path (repository; common prefix `TanitAD Research Hub/Architecture & Inference/Research/`)
contrastive-bias toy, first version (defective)	`reff_pmi_toy.py`, `reff_pmi_toy_result.json`
nested-vocabulary coverage	`reff_v0_coverage.py`, `reff_v0_coverage_result.json`
kinematic mask	`2026-09-29-hicap/hicap_prior_mask_measure.py`, `hicap_prior_mask_result.json`
conformal keep-set, real windows	`2026-09-29-hicap/hicap_conformal_cells_measure.py`, `hicap_conformal_cells_result.json`
imagination toy	`2026-09-29-hicap/rc_latent_imagination_toy.py`, `rc_latent_imagination_toy_result.json`
hierarchical retrieval toy	`2026-09-29-hicap/rb_hier_retrieval_toy.py`, `rb_hier_retrieval_toy_result.json`
camera-gating toy	`2026-09-29-hicap/rd_camera_gating_toy.py`, `rd_camera_gating_toy_result.json`
contrastive-bias toy, corrected	`2026-09-29-hicap/hicap_pmi_toy_v2.py`, `hicap_pmi_toy_v2_result.json`
red-team checks (masked InfoNCE, conformal)	`2026-09-29-hicap/RT_masked_infonce_check.py`, `RT_conformal_below_nominal.py`
cost model (arithmetic)	`2026-09-29-hicap/hicap_cost_model.py`, `hicap_cost_model_result.json`

10 Pre-registered evaluation protocol

This protocol is committed before any HiCAP checkpoint exists. It extends the programme's draft pre-registration for the flat contrastive selector (REF-F, hypotheses H-F1–H-F8), which it contains as the special case of one level, one backbone and no imagination. The freeze rule is the programme's: the PI approves the document, its hash is recorded before the first checkpoint, and any later change is a dated amendment.

Data, estimator, reference. Training uses the v7 corpus (4,572 clips); evaluation uses the programme's T1 open-loop panel of 4,823 windows over 141 episodes. The corpus is non-parity, so results are valid as

differences between arms trained on it and never as levels against parity arms. Every interval is an episode-cluster bootstrap (B = 2000, cluster = episode) on the full-set mean; two arms on the same windows are compared with the paired version; the deprecated split-mean estimator is refused. Because a zero-lever replicate has been measured to read "separably worse" on 5 of 9 family metrics in this programme[I], every arm carries a seed-1 replicate and a bar is met only if the effect is at least 10% relative and at least three times the replicate floor measured in the same panel. The reference arm is refcv5-v2 (the newest arm with results; the base configuration of the never-launched refcv6), with the hold-current-action control `ha`, its constant-velocity extension `ha0_ext` and `ha0`; any HiCAP arm must beat `ha0_ext` and `ha` separated, not just the trained reference. Learning-curve exponents are not used: data-efficiency claims use matched-step ratios only.

Four metric families, per family, never pooled. *Longitudinal*: speed MAE, target-speed accuracy at 0.5 m/s, along-track MAE, headway, time-gap and minimum TTC with n_{closing} and the censored share. *Lateral*: heading, yaw rate, cross-track and *masked* curvature against the straight-line floor. *Tactical*: lateral and longitudinal accuracy and κ of the factored decision, cell-selection accuracy, goal-point error and bearing. *Strategic*: route accuracy, κ and n; reported UNAVAILABLE with the reason where $n = 0$ (today five strategic actions are usable, §4.2). Rows are stamped: T1 open loop, non-parity, and for every nav-fed row the paired nav-zero row, since the navigation command is an oracle input. Strategic verdicts are read on the nav-zero rows (nav-fed route accuracy is partly an echo of the input). Every panel also reports the ego-zeroed (image-only) row for every family, not only as a control.

Negative controls, run first; a panel is void if one fails in the stated way. **NC1** state permutation (embeddings shuffled across windows): ADE must degrade to at least the frequency-prior baseline. **NC2** random vocabulary instead of farthest-point codes: must not beat the designed vocabulary. **NC3** k-NN retrieval in the same frozen feature space: HiCAP must be separated-better, else the heads add nothing. **NC4** ego-only and image-only variants: the ego-only score must not approach the full model (copycat risk). **NC5** nav-zero and nav-shuffled. **NC6** flat tactical (no parent restriction), shuffled parent, a goal-swap causality test and a strategic-token-null ablation (attribution between the goal path and the shared trunk). **NC7** imagination-shuffled and action-shuffled. **NC8** random camera/token gate at the matched cost, plus (for any gate that reads the previous tactical token) the config-invariance and gate-shuffle tests. **NC9** audio-null (bit-exact), audio-shuffled, bearing flip, level dose-response, onset shift, and the audio-only label probe.

Table 16: Hypotheses with both outcomes committed. "Non-inferior" means the upper 95% bound of the paired difference in the family's primary metric lies below its pre-set margin δ_f (for ADE@2 s, +0.02 m; the other margins are set before launch from the replicate floor) and is required in every family; "separated" means the 95% interval excludes zero. Thresholds are pre-registered proposals that the PI may amend before the freeze. Verdicts are SUCCESS, FAIL, INCONCLUSIVE, MISSING_DATA or NOT_TRIGGERED; INCONCLUSIVE and NOT_TRIGGERED are not passes.

id	measurement	CONFIRM if ⇒ consequence	REFUTE if ⇒ consequence
H-HC0 gate	k-NN over frozen VLM tokens, motion tokens and v_0 versus hold-speed *and versus a v_0-only k-NN*, non-steady windows, paired, before any head is trained	frozen+motion+v_0 k-NN separated-better than the v_0-only k-NN ⇒ the frozen features carry decision signal beyond the ego state; proceed	not separated ⇒ change backbone or adapter before any head training
H-HC1 adapter	paired ADE with the ego channel zeroed, with versus without the motion adapter	with-adapter separated-better by ≥ 10% ⇒ adapter is needed for metric motion	not separated ⇒ drop the adapter (2 M parameters)
H-HC2 driving knowledge	Alpamayo-1.5 language model versus its parent Cosmos-Reason2-8B (same architecture, same heads; 8 B against 8 B, not size-matched to the primary), paired, all four families, on clips passing the 147-clip contamination screen; second surface off PhysicalAI; Qwen-Drive-1.0-4B versus Qwen3.5-4B as an external 4 B replication	the driving-tuned model separated-better on ≥ 1 family, non-inferior on the rest, on both surfaces ⇒ driving tuning helps a frozen interface	the parent non-inferior ⇒ keep the general backbone; a gain only on PhysicalAI is reported as contamination-suspect (no arm is a clean control, §8.6)

id	measurement	CONFIRM if ⇒ consequence	REFUTE if ⇒ consequence
H-HC3 tree vs flat	beam-retrieved tree and factored 20×10 compose versus flat softmax over the identical 262,144 entries; agreement with the flat pick; oracle-in-composed versus oracle-in-flat	non-inferior, composed oracle within 5% of flat, $\geq 10\times$ fewer rows ⇒ adopt the tree (justified by the small set for the backbone and by priors, not by speed: flat costs ≈ 1 ms)	lower bound $> +0.02$ m or composed oracle loss $> 5\%$ ⇒ raise (k_P, k_V) or add a joint cell stage; else ship flat plus pruning
H-HC4 prior correction	KL of selected-class frequency to ground truth, and steady-window ADE: masked InfoNCE, unmasked InfoNCE and their β-corrected versions versus soft-target softmax	InfoNCE-only has the larger KL and separated-worse steady ADE ⇒ correction mandatory	no separated difference ⇒ drop the correction
H-HC5 vocabulary scale	CPU first (E-R2): product vocabularies at $N \in \{2^{10}, 2^{12}, 2^{14}, 2^{16}, 2^{18}\}$ from the training split only, oracle-in-set ADE on validation, matched-N ratios (no exponent); then tree-selected and flat-selected ADE at each size	$4\times$ entries ($2^{14} \to 2^{16}$) still cut oracle ADE $\geq 15\%$, and tree-selected non-increasing to 2^{16} while flat-selected worsens (paired, separated) ⇒ keep 262,144, consider 1.05 M	$< 5\%$ per $4\times$ beyond 2^{12} ⇒ stop at ≤ 16 k entries: the large vocabulary is a prior structure, not a size
H-HC6 pruning	R1 retention of logged futures; conformal coverage on episode-disjoint test, *per tactical class*; share of nodes removed; survival versus survivor-cache age $0 \ldots 4$ ticks; agreement of cells minted over $[2, 6]$ s with the shipped labels	retention lower bound ≥ 0.99; mean coverage over ≥ 200 episode-disjoint calibration draws $\geq 1 - \sum \alpha_\ell - 0.01$ and the 5th-percentile draw $\geq 1 - \sum \alpha_\ell - 0.05$; survival $\geq 99\%$ in every class with $\leq 60\%$ kept; age-4 survival within 0.5 points; cell agreement ≥ 0.9 ⇒ rules ship as hard masks	any class $< 97\%$ ⇒ demote that rule to a soft penalty (keep physical limits only); cache age loses > 2 points ⇒ cadence 2–3 ticks; agreement < 0.7 ⇒ 6 s candidates required
H-HC7 re-score	candidate-conditioned late-interaction re-score versus dot product only	separated-better on ADE or on the longitudinal family, non-inferior on the rest ⇒ keep the block	not separated ⇒ the dual encoder suffices; drop it
H-HC8 imagination	first E-I0: probes from the frozen state to lead gap and closing speed versus a v_0-only baseline and a shuffled cue; then Tier 0 (soft targets) versus one-hot; Tier 1 on top of Tier 0; Tier 2 veto; headway and TTC on closing windows; Spearman(confidence, regret) with and without each term; always the risk-sensitive floor and a model-free value on the same rows	E-I0 exogenous $R^2 \geq 0.3$ above the baseline with CI excluding 0; Tier 0 lowers the selection gap by $\geq 15\%$ (paired); Tier 1 separated-better longitudinally beyond Tier 0; NC7 removes $\geq$ half the gain; confidence sign stays negative ⇒ keep the tiers that pass	E-I0 $R^2 < 0.1$ ⇒ Tier 1 is dead for this encoder, ship Tier 0 only and reopen the encoder question; no Tier 0 effect ⇒ drop it; sign flips ⇒ add a calibrator
H-HC9 student (conditional)	run only if H-HC11 excludes every backbone tier at the needed camera count: HiCAP-S versus HiCAP-R, paired, four families, level-2 agreement, share of windows meeting the margin condition	non-inferior in every family, agreement $\geq 90\%$, condition met on $\geq 80\%$ ⇒ the student is admissible at the failed tier, with the guarantee	otherwise ⇒ end-to-end student, guarantee lost. If no tier failed the verdict is NOT_TRIGGERED, which is not a pass
H-HC10 elastic sensing	*first* E-C3: oracle-camera value by manoeuvre on a ∼30-chunk side-study ($n \geq 30$ per stratum), paired, four families; then gate G-V versus fixed-all at the peripheral budget and versus a random gate at matched cost; G-V per-class recall and wrong-run length	oracle camera beats front-only beyond the harm margin in ≥ 1 stratum, and G-V has paired CI excluding the harm margin in every family, mean cost $\leq 0.35\times$ fixed-all, p95 ≤ 100 ms, separated-better than random, wrong runs $\lesssim 3$ ticks ⇒ adopt G-V and tier only those strata	no stratum where an oracle camera helps ⇒ retire elasticity, ship front-only (a compute win, no caches bought); a weak G-V ⇒ PI ruling on ego cues (G-E) or hard speed/route tier rules. INCOMPLETE for cameras until caches exist

id	measurement	CONFIRM if ⇒ consequence	REFUTE if ⇒ consequence
H-HC11 latency, contention and precision (per backbone tier)	$\geq$ 1,000 refresh ticks and $\geq$ 1,000 fast-path ticks on Thor for each tier of the ladder: fast-path p95 alone and *concurrent with a running refresh* (high-priority stream), refresh T_r p95, duty $D = f_r(T_r + o)$ at the chosen cameras and f_r, token age p95, L_{cmd} p95, peak memory from `max_memory_allocated` only, NVFP4 versus BF16 pooled-feature drift on real frames	fast path p95 concurrent ≤ 25 ms, L_{cmd} p95 ≤ 100 ms, $D \leq 0.5$, drift within the margin fixed beforehand ⇒ the tier is admitted; the the primary (Cosmos3-Edge) is admitted or excluded by these numbers like any tier, and a larger tier replaces it only if H-HC16 shows a gain (no parameter count is consulted)	any criterion failed ⇒ that tier is excluded by that number; contention alone failing ⇒ lower f_r or partition the SoC; no tier of ≥ 2 B admitted ⇒ H-HC9 is triggered
H-HC12 audio	T1 audio-null equals the original embedding bit-for-bit; T2 event heads (class, bearing, distance) on injected events; T3 controls (null, shuffle, bearing flip, dose-response, onset shift, no-vehicle/no-siren, audio-only label probe); T4 no regression on the four families	T1 exact, T2 above baseline, flips and dose-response behave, audio-only probe at chance, T4 non-inferior ⇒ the path is wired and controlled (*not* a driving-skill claim; behaviour only in a closed-loop acoustic simulation)	probe above chance ⇒ the generator leaks the label, regenerate from world state; controls fail ⇒ shortcut, audio path unusable here
H-HC13 data efficiency	four-family panel at 3/10/30/100% of training hours, nested episode subsets, versus the reference at matched fractions	relative ADE loss at 10% and 3% separated-smaller than the reference's ⇒ first evidence for the data goal	not separated ⇒ no data-efficiency edge here
H-HC14 closed loop	NuRec/AlpaSim, paired against the reference: pass rate, collisions, jerk, $n \geq 30$ scenes	pass rate non-inferior, jerk not separated-worse ⇒ closed-loop reference	separated-worse ⇒ strengthen stage 3. $n < 30$ gives INCOMPLETE regardless of the point estimate (the programme's provisioned scene count is currently 1)
H-HC15 token age	fast path fed cached VLM tokens aged 0/0.5/1.0/1.5 s, (a) trained with age jitter as designed and (b) trained at age 0 and tested aged; four families, paired on the same windows; E-I0 probe R^2 against age	age 1.0 s non-inferior to age 0 in every family ⇒ the 2 Hz asynchronous refresh is admissible and the age cost of a larger tier is affordable	age 1.0 s separated-worse in a family ⇒ raise f_r or restrict the backbone to the tier whose worst-case age is inside the measured tolerance; the tolerance curve, not a parameter count, becomes the size limit
H-HC16 backbone scale ladder	same heads on cached features from Cosmos-Reason2-2B, Qwen3-VL-4B (the intermediate rung; no Cosmos 4 B reasoner exists) and Cosmos-Reason2-8B, once their identity with the Qwen3-VL architecture is verified (three points: no exponent), and Cosmos3-Edge as an off-architecture point; matched cameras and f_r, four families, paired, matched-step ratios	the 8 B tier separated-better than the 2 B tier in $\geq$ 1 family and non-inferior in the rest, inside an admitted H-HC11 tier ⇒ scale earns its latency; continue only through a measured tier	no separated gain of the 8 B tier over the 4 B rung, or of the 4 B rung over the 2 B tier ⇒ the frozen state saturates at the first rung without gain (reported per step, not per pair of ends); Cosmos3-Edge, the primary, is placed by its own paired difference to the 2 B tier and the 8 B tier; stop scaling and spend on adaptation (H-HC17) or on data
H-HC17 partial adaptation	low-rank adapters on the language model's upper layers and on the vision tower of the H-HC16 winner ($\approx$ 8–13 GPU-hours per epoch, [E]), paired against the frozen arm, four families, ego-zeroed row, NC1, contamination-screened clips only; run only after H-HC0 and H-HC16	separated-better than frozen on a metric family (speed MAE or along-track) and non-inferior in the rest ⇒ adapt, accepting the loss of the feature cache	not separated ⇒ stay frozen and cached; the motion adapter (H-HC1) is the geometry fix

Reading "non-inferior in every family". The strategic family currently has five usable actions and too few windows for any interval; for H-HC9, H-HC15, H-HC16 and H-HC17 the clause is evaluated on the families with $n \geq 30$ windows and reported UNAVAILABLE, with n, for the others, never silently passed. Margins δ_f are numeric and fixed before the first checkpoint from the replicate floor measured in the same panel (ADE@2 s: +0.02 m; speed MAE, along-track, heading, yaw rate and cell accuracy: the larger of 10% relative and three replicate standard deviations).

Stopping rules. If NC1 fails, stop and diagnose before reading any other result. If H-HC0 is refuted, no head is trained on that backbone. No arm is tuned on validation; temperature, β, λ, α_ℓ and the gate thresholds are fitted on held-out training episodes.

What this protocol does not test. It is not a strategic-level result (five usable strategic actions, no route supplier other than the ego's own future, no map); it is not a safety claim (pruning that uses agent states is an oracle-perception simulation until a learned range head exists); and it says nothing about the released CLM-8B model itself.

11 Differentiation from known approaches

Table 17 compares HiCAP's design with the closest published families on nine properties. The table is our reading of each system's main configuration and errs toward crediting others (a partial mark where a property is present in weaker form). A cell for HiCAP means *specified in this paper*, not demonstrated: no HiCAP model has been trained.

Table 17: Property matrix. ✓ present; ∘ partial or weaker form; — absent; · not applicable (the property concerns a contrastive score, and the system uses a softmax classifier). Columns: (1) frozen large backbone; (2) no autoregressive decode at run time; (3) large fixed action vocabulary; (4) hierarchical tree retrieval; (5) correction of the prior bias of the contrastive optimum; (6) pruning with a stated recall guarantee; (7) outcome or cost prediction per candidate; (8) elastic choice of sensors or tokens; (9) audio tokenised into the backbone space.

system	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
EMMA, SimLingo, Alpamayo-R1 [14, 28, 33]	∘	∘	—	—	·	—	—	—	—
DriveVLM-Dual, FASIONAD [31, 35]	—	∘	—	—	·	—	—	—	—
DiffusionDrive [25]	—	✓	∘	—	·	—	—	—	—
VADv2 [6]	—	✓	✓	—	·	—	—	—	—
Hydra-MDP [22]	—	✓	✓	—	·	—	✓	—	—
GTRS, DriveSuprim, SparseDriveV2 [24, 34, 42]	—	✓	✓	∘	·	—	✓	—	—
WoTE, ZTRS [20, 23]	—	✓	✓	—	·	—	✓	—	—
IBC [10]	—	✓	—	—	·	—	—	—	—
CLM [18]	✓	✓	∘	—	—	—	—	—	—
CoVer-VLA [19]	✓†	✓	—	—	—	—	—	—	—
HiCAP (this design)‡	(✓)	(✓)	(✓)	(✓)	(✓)	(✓)	(✓)	(✓)	(✓)

† a frozen image encoder, not a full vision–language model. ‡ parenthesised marks mean specified, not demonstrated; the others mean shown by the cited system.

What is new, and what is not. Frozen-backbone contrastive selection (CLM, CoVer-VLA), large fixed vocabularies with per-candidate cost heads (VADv2, Hydra-MDP, GTRS) and coarse-to-fine scoring (DriveSuprim) are prior art; so are latent world models and conformal prediction. Tree retrieval over node embeddings, cascades of conformal sets with a summed miss budget [9], predict-then-plan with an open-loop forecast and an analytic time-to-collision cost, and outcome critics of the Contrastive RL kind are likewise known. The claim of this paper is the *composition* and its application to driving under this programme's constraints, together with four points whose *combination* we did not find in the literature we read (absence at the depth of our search, not a proof):

1. **A frozen, versioned, hierarchical token vocabulary as the action side of a contrastive selector**, with text-defined nodes, a factored path-by-profile trajectory vocabulary beneath them (the structure of Sun et al. [34], the densest we found, here partitioned by the programme's lateral and longitudinal tokens), level-wide marginal critics, cost and bandwidth that scale with the number of rows touched rather than entries (Proposition 6), and factored lateral/longitudinal scoring so that the two decisions remain separately measurable.
2. **A pruning cascade with a stated recall bound whose survivors return to the backbone.** Physics, state-conditioned admissibility, hysteresis and conformal keep-sets reduce the set before any expensive step (Proposition 7, measured mask in §9), and the survivors become candidate tokens for late-interaction re-scoring and an optional language verifier. Coarse-to-fine scorers prune with learned scores and no coverage statement.
3. **Identifiable, tiered imagination.** Restricting imagination to outcome-aware soft targets, an analytically integrated forecast of non-reactive exogenous descriptors and a gated critic with a disagreement

veto makes the counterfactual evaluation of a candidate a statement about observational data (Proposition 9); our own toy and earlier measurements showed why an unrestricted learned counterfactual is unsafe (anti-conservative without counterfactual labels, and it inverts the sign of confidence against regret).

4. **Elastic sensing and audio under one embedding interface with a distillable state side.** Camera and token budgets are chosen per tick from cheap cues, audio (optional at inference) enters the backbone's token space with explicit null and shuffle controls, and because action embeddings never depend on images, a small student can occupy the same interface and reuse the teacher's action table, with a decision-preservation margin (Proposition 11) should a latency tier fail.

Relation to the programme's own designs. Two internal designs overlap with HiCAP, and it is stated here so that neither is re-invented. The flat contrastive selector (REF-F, pre-registration H-F1–H-F8) is the one-level special case of HiCAP, and its hypotheses are contained in §10. TanitLang's "Energy Bridge" design (the programme's latent-reasoning proposal, dated 2026-09-06) re-ranks the planner's fan by a learned compatibility energy $S_1(k) = S_0(k) - \beta\, E_\phi(z, \tau_k)$ and audits consistency with the same function; language is kept off the control path, and its released-vocabulary layer enters as zero-initialised log-priors on the anchors[I]. HiCAP can be read as the missing *objective* for that energy: with $E = -e^\tau \langle \hat{\mathbf{s}}, \hat{\mathbf{a}} \rangle$ the authority β and the logit scale are one degree of freedom, and InfoNCE is the contrastive answer to the design's open question of how the energy should be trained. The differences are that HiCAP's vocabulary nodes are *actions* carrying trajectory codes rather than priors on a fixed fan of 117–128 anchors, that the selector may be standalone (full authority) rather than a re-ranker, and that the model-selected positive of the consistency loss and the ground-truth-nearest positive of the selector loss are kept as separate terms. HiCAP inherits TanitLang's constraint that language is not on the control path, and its controls (constant, shuffled and norm-matched state) reappear as NC1. The symbol β means authority there and the log-prior weight here; we use β_ℓ only for the latter.

Where HiCAP is deliberately conservative. Language never generates: it embeds token definitions and, only in uncertain ticks, verifies. Nothing in the goal path carries a situation-classifier output. Every oracle input is stamped and paired with its ablation. And hypotheses are registered with their refutation consequences, so that a negative outcome (for instance, that a larger vocabulary does not help, H-HC5) redirects the programme instead of being explained away.

12 Limitations, open decisions and conclusion

What this paper does not show.

1. **No HiCAP model has been trained.** The evidence is: proofs of elementary statements, real-data measurements that use only inference-time quantities (a kinematic mask, nested-vocabulary coverage and a conformal keep-set; §9), toy simulations whose properties belong to synthetic worlds, closed-form arithmetic and NVIDIA-published backbone timings. The cost tables are not timings of HiCAP, and the fast path has never run on the target SoC.
2. **The supervision is thinner at the top of the hierarchy than the thesis needs.** Five strategic actions are usable; lane changes are unreachable as tactical actions; eight of the 22 tactical goals have fewer than 30 examples; the perception-style goals are unverified teacher text, and a 22-token goal head in the programme's newest trained arm received exactly zero gradient for all 40,284 steps, so *no measured prediction quality of any v7 token head exists*[I]. HiCAP would be the first arm to measure it.
3. **Oracle and privileged channels.** The navigation command is the ego's own future turn, admitted under a programme ruling and stamped; the maximum-speed input is an echo of the ego's realised speed and is not used. Perception labels come from a teacher. All are paired with controls, but none of the controls has been run.
4. **Assumptions inside the theory.** Identifiable imagination assumes a non-reactive scene over the horizon; conformal guarantees are marginal and hold across, not within, episodes; the physical-limit mask was checked on 0.5 s finite differences of four waypoints, coarse but conservative, on 881 windows from 40 episodes.
5. **The latency criterion is arithmetic, not a measurement.** The PI's size ruling (2026-09-29) replaced the sub-300 M cap by a requirement that the whole system's inference time be very low. Tables 6 and 7 are closed-form scalings of NVIDIA-published Qwen3-VL rows on Thor; no published row exists for Cosmos-Reason2, the Cosmos 3 Nano reasoner, the Alpamayo language model or an omni model, refresh-versus-fast-path contention has not been observed, the cost of stale tokens is unmeasured, and the assumptions (duty ≤ 0.5, a 15 ms overhead, a 64 GB share of Thor's memory) are ours. The student, whose guarantee

needs a score margin that may not exist on real windows, is only a fallback (H-HC9, H-HC11, H-HC15).

6. **Data the design needs does not exist yet.** Only the front-wide camera is cached (elastic sensing cannot be evaluated on real cameras); no corpus the programme uses carries audio; and the validation set that would judge a driving-aware backbone is contaminated to an extent we could bound only for a related split.
7. **Integration state.** The vocabulary module, the label consumer and the reference planner code live on the PI's unpushed branch and in his Drive, not on any GitHub branch; the reference arm refcv6 was pre-registered but never launched; and the document referred to as "refcv7" does not exist (the nearest is refcv6). The earlier REF-F plan's integration section was written against a repository state that is now stale.

Conflicts between programme documents, reported not resolved. The registry and a later architecture review disagree on whether the refined-confidence selection lever inherited by every refcv6 arm is proven or a measured negative (+0.0259 m, separated)[I]; and the route accuracy credited to a 771-parameter head appears to come from the core's nav-blind route probe[I]. Both bear on any claim about selection.

Decisions the PI has made (2026-09-29). (a) A VLM is the preferred backbone, drawn from NVIDIA's Cosmos family (the repository the PI named). (b) Audio may be used optionally at inference. (c) The sub-300 M cap does not bind if the inference time of the whole system is very low; §8.3 turns that condition into three pre-registered criteria.

Further rulings (2026-10-25). The latency assumptions are accepted (duty ≤ 0.5, a 64 GB memory share, L_{cmd} p95 ≤ 100 ms); longitudinal acceleration and yaw rate are admitted as ego inputs, curvature is not; no Thor measurement starts before the plan is complete; the primary backbone is the 4 B Cosmos3-Edge checkpoint, with Cosmos 3 Nano as the alternative; training uses the programme corpus with the newest tactical-label release (§3.1).

Decisions that still belong to the PI. (1) Are ego cues admissible for the camera gate (a gate is a situation classifier in disguise, and a vision-only gate measured much weaker in the toy)? (3) Recover the 40 exact validation clip identifiers so that contamination can be tested; (4) which branch to build the reference code on, given that main is stale and the newest code is unpushed; (5) fund the ≈ 0.1 GPU-hour probe E-I0 that decides whether the frozen state carries metric gap and closing speed, before any imagination work, and the on-Thor run of the H-HC11 benchmark, which is the first measurement of any HiCAP-relevant latency; (6) approve or amend the thresholds in Table 16 before the freeze; (7) approve the one-off cost of the adapted-backbone arm (H-HC17, $\approx$ 8–20 GPU-hours per epoch) only after H-HC0 and H-HC16.

Conclusion. The programme's measured bottleneck is the decision, not the candidate set. HiCAP proposes to build the decision maker on a backbone that already knows the world, using the machinery that made contrastive selection cheap in language models, and to add what driving needs and language does not: a hierarchy over the programme's own frozen vocabulary, pruning with a stated recall, imagination restricted to what data can identify, sensors and tokens bought by value, and a latency rule, not a parameter cap, for how large the backbone may be. The paper commits the tests, including the ones that would sink parts of it. The first of them (H-HC0) is cheap and decides whether the rest is worth building.

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A The v7 vocabulary as used by HiCAP

Counts are per field over the 4,719 clips of the v7 corpus (label census of the v8 release)[1]; `g_tac` is multi-label, so its counts do not sum to 4,719. "geometry" tokens derive from the ego's own future path, "Alpamayo text" from the teacher's chain-of-causation text (unverified; 49.6% of records carry at least one disputed goal). Empty tokens are masked in the loss and in retrieval. Token ids are tuple indices in the frozen vocabulary; `REDUCE_TO_FOLLOW_ROUTE` is *not* part of it. The 22 exclusive goal pairs and the goal-to-action admissibility tables are part of the same frozen module.

field	id	token	n	derivation	note
g_{str} (8)	0	`FOLLOW_ROUTE`	3,041	geometry	
	1	`TURN_LEFT_FOLLOW_ROUTE`	585	geometry	
	2	`TURN_RIGHT_FOLLOW_ROUTE`	624	geometry	
	3	`STOP_AT_FOLLOW_ROUTE`	469	geometry	
	4	`EXIT_LEFT_FOLLOW_ROUTE`	0	geometry	empty
	5	`EXIT_RIGHT_FOLLOW_ROUTE`	0	geometry	empty
	6	`LANE_CHANGE_L_FOLLOW_ROUTE`	0	geometry	empty
	7	`LANE_CHANGE_R_FOLLOW_ROUTE`	0	geometry	empty
a_{str} (7)	0	`HOLD_MAIN_ROAD`	2,468	geometry	
	1	`PREPARE_TURN_L_FOLLOW_ROUTE`	585	geometry	
	2	`PREPARE_TURN_R_FOLLOW_ROUTE`	624	geometry	
	3	`PREPARE_STOP_FOLLOW_ROUTE`	469	geometry	
	4	`PREPARE_EXIT_FOLLOW_ROUTE`	0	geometry	empty
	5	`PREPARE_LANE_CHANGE_FOLLOW_ROUTE`	0	geometry	empty
	6	`RESUME_CRUISE_FOLLOW_ROUTE`	573	geometry	
$a_{\text{tac}}^{\text{lat}}$ (8)	0	`LANE_KEEP`	3,057	geometry	
	1	`LANE_CHANGE_L`	0	geometry	empty
	2	`LANE_CHANGE_R`	0	geometry	empty
	3	`ABORT_LC`	0	geometry	empty
	4	`NUDGE_L`	502	geometry	
	5	`NUDGE_R`	613	geometry	
	6	`TURN_L`	280	geometry	
	7	`TURN_R`	267	geometry	
$a_{\text{tac}}^{\text{lon}}$ (8)	0	`FOLLOW`	175	geometry	
	1	`CRUISE`	1,287	geometry	
	2	`YIELD_MERGE`	0	geometry	empty
	3	`BRAKE_TO`	927	geometry	
	4	`CREEP`	141	geometry	
	5	`HOLD`	133	geometry	

field	id	token	n	derivation	note
	6	ADAPT_SPEED_FOR_CURVE	1,018	geometry	
	7	ACCELERATE	1,038	geometry	
g_{tac} (22)	0	FOLLOW_LANE	3,748	geometry	
	1	TURN_L	280	geometry	
	2	TURN_R	267	geometry	
	3	YIELD_FOR_TURN_L	23	Alpamayo text	$n < 30$
	4	YIELD_FOR_TURN_R	20	Alpamayo text	$n < 30$
	5	YIELD	626	Alpamayo text	
	6	STOP_POINT	336	geometry	
	7	SPEED_BAND	4,719	geometry	
	8	CORRIDOR_OFFSET	885	Alpamayo text	
	9	EVADE_IN_CORRIDOR	246	text $\wedge$ geometry	
	10	OVERTAKE_VEHICLE	21	Alpamayo text	$n < 30$
	11	MERGE	84	Alpamayo text	
	12	GAP_TARGET	382	Alpamayo text	
	13	REACT_ON_ONCOMING	344	Alpamayo text	
	14	TAKE_EXIT_L	21	Alpamayo text	$n < 30$
	15	TAKE_EXIT_R	133	Alpamayo text	
	16	TRAFFIC_LIGHT_REACT	18	Alpamayo text	$n < 30$
	17	TRAFFIC_LIGHT_REACT_RED	384	Alpamayo text	
	18	TRAFFIC_LIGHT_REACT_YELLOW	25	Alpamayo text	$n < 30$
	19	TRAFFIC_LIGHT_REACT_GREEN	378	Alpamayo text	
	20	LANE_CHANGE_L	23	Alpamayo text	$n < 30$
	21	LANE_CHANGE_R	16	Alpamayo text	$n < 30$
nav_command (3)	0	NAV_FOLLOW_ROAD	2,993	ego-future	
	1	NAV_TURN_L	824	ego-future	
	2	NAV_TURN_R	902	ego-future	